

## SURVEY ARTICLE OPEN ACCESS

## The Monetary Policy–Commodities Nexus: A Survey

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## ABSTRACT

This survey synthesizes evidence on the bidirectional links between commodity markets and monetary policy. On the commodities-to-policy side, we review how shocks to energy, food, and metals pass through to inflation, inflation expectations, economic activity, and financial stability in state-dependent ways that vary by shock type, exposure, and policy regime. We complement the literature with an analysis of central-bank speeches, showing how officials classify commodity shocks and how these framings map into policy stances. On the policy-to-commodities side, we organize evidence on the transmission of monetary policy to commodity markets via financial, real-economy, and expectations channels, highlighting heterogeneity across policy instruments, commodities, and central banks. We emphasize how financialization tightens cross-asset linkages, raises leverage and margin sensitivity, and amplifies discount-rate and risk-taking mechanisms. Overall, commodities are best treated as policy-sensitive state variables, not exogenous disturbances, with implications for policy design, central bank communication, and international monetary spillovers.

**JEL Classification:** E31, E32, E44, E52, Q02

## 1 | Introduction

Over recent decades, the global economy has weathered an extraordinary sequence of large and persistent shocks. Among these disruptions, commodity price shocks and volatility have proved influential in shaping inflation, inflation expectations, the business cycle, and financial stability. Yet there remains a fundamental debate over the nature of such commodity shocks and the channels through which they operate. This survey is organized around an ongoing dilemma, namely, to what extent commodity shocks are external disturbances that monetary policy should absorb, and to what extent they are endogenously shaped by monetary conditions. This distinction bears directly on policy design, guiding both how information on commodity prices should enter the reaction function of central banks as well as which role commodities play in the monetary transmission mechanism (IMF, 2024; Gagliardone and

Gertler, 2023; Dao et al., 2024; Bernanke and Blanchard, 2025; Avalos et al., 2025).

Historically, the prevailing view defaulted to a supply-shock reading of commodity price pressures, shaped by the 1970s oil shocks. That episode entrenched the view that commodity shocks raise costs and subsequently inflation while weighing on real activity. In the ensuing decades, monetary policy took primacy in safeguarding low and stable inflation, with economic theory underscoring the central role of managing inflation expectations. Consistent with that logic, central banks often “look through” the first-round effects of commodity price shocks, particularly when these are expected to be temporary. Once second-round effects via wages or markups are evident, and the risks of de-anchoring expectations rise, central banks typically tighten. This operational approach reflects the long-standing tradition of modeling commodity shocks as exogenous relative-price shocks

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and conditions the policy response on estimated persistence and pass-through (e.g., Lane, 2023; Ueda, 2023; Mann, 2023).

The mid-2000s commodity super-cycle brought this trade-off into sharp focus, even though policymakers were alert to it from the outset (Blanchard and Gali, 2007; Bernanke, 2008; Kilian, 2009; Baumeister and Peersman, 2013). On the eve of the Great Financial Crisis (GFC), then Bank of Canada Governor Carney, for example, emphasized how substantial Asian demand, especially from China, was propelling commodity prices, framing it as a durable relative-price shift to be looked through absent second-round pressures (Carney, 2008). Around the same time, in July 2008, the European Central Bank (ECB) raised rates to safeguard medium-term price stability and contain second-round risks (Trichet, 2008). This decision was later criticized as financial conditions deteriorated. Taken together, these episodes expose a persistent identification and policy-design problem. When the persistence and pass-through of relative-price shocks are uncertain in real time, the choice between tightening to safeguard expectations and accommodating to support economic activity remains contested (e.g., Kuroda, 2022; Lane, 2023; Cunliffe, 2022).

Commodities, however, should not merely inform the design of policies, but linkages run in both directions, with policies shaping prices and market liquidity. In particular, we systematize evidence showing that monetary policy shocks affect commodity prices through three sets of channels—financial, real-economy, and expectations-based—with effects that emerge over horizons from minutes to weeks, depending on the channel (e.g., Anzuini et al., 2013; Hammoudeh et al., 2015; Filardo et al., 2020; Apergis et al., 2020; Yu et al., 2023; Degasperis et al., 2024; Miranda-Pinto et al., 2025). Empirically, responses to monetary policy vary markedly in timing, magnitude, and even sign across commodities, policy shocks, and regimes. As such, treating “the commodity market” as a single object, or using a single commodity like oil as a proxy, can substantially bias inference. Moreover, these results clearly show that commodity markets are an integral part of the monetary transmission mechanism. That is, they not only signal macroeconomic conditions relevant to policymakers but also propagate monetary policy shocks.

The financialization of commodity markets, marked by the growing presence of (often passive) financial investors, deeper derivatives activity, and tighter links to wider asset markets (Cheng and Xiong, 2014; Adams et al., 2020; Bianchi et al., 2020), has expanded and deepened the connection between monetary policy and commodity markets. On the one hand, financialization alters the propagation of commodity shocks via funding constraints, margining practices, and common risk factors, thereby amplifying or dampening pass-through to inflation and economic activity. On the other hand, it heightens commodities’ sensitivity to monetary impulses transmitted through discount rate, risk-taking, exchange rate, and cross-asset flows. Moreover, because commodities now sit within leveraged, multi-asset portfolios and common-creditor networks, disturbances can trigger margin spirals and forced deleveraging, raising risks to financial stability (Apostolakis et al., 2021; Daskalaki and Skiadopoulos, 2016).

The current confluence of rising geopolitical risks and continuing fragmentation of global commodity markets, as well as ongoing reviews of central bank policy frameworks at major central banks provides an opportune moment to reassess the channels linking monetary policy and commodity markets. Against this backdrop, this paper reviews several key questions. How should commodity markets inform monetary and financial-stability policy? What determines the magnitude and persistence of commodity shock pass-through into the broader macroeconomy? How do central bankers communicate the role of commodity developments in policy decisions? How can we best understand commodity price fluctuations that reflect both external shocks and endogenous responses to monetary and macro-financial conditions? We address these questions through a narrative, critical survey with an extensive and wide-ranging selection of literature, prioritizing recent high-impact and foundational studies. Specifically, we focus on studies that (i) quantify the commodities-to-macro pass-through, (ii) identify the policy-to-commodities transmission, and (iii) speak to the effects of financialization and central bank communication across commodities and central banks.

Our contribution is twofold. First, we synthesize evidence on how commodity-price shocks affect inflation, economic activity, and financial stability. We show that the magnitude and persistence of the macroeconomic effects of commodity-price shocks are state contingent, with implications for distinguishing temporary from persistent episodes and for pairing interest-rate policy with macroprudential tools. To support this narrative, we also analyze a corpus of recent speeches by central bankers, tracing how officials frame and classify commodity shocks across central banks and how those framings enter their respective policy stances. Second, we synthesize evidence that monetary policy moves commodity prices via financial, real-economy, and expectations channels, and we organize the heterogeneity in responses across policy instruments, commodities, and central banks. Throughout, we treat financialization and policy communication as cross-cutting modifiers of both the commodities-to-policy and policy-to-commodities linkages. Taken together, the evidence in this survey calls for treating commodities as central, policy-sensitive state variables rather than peripheral disturbances in both the design and transmission of monetary policies.

In light of the foregoing, this survey connects two strands of literature. In the next two sections, we first review the macro-commodity interactions literature, exploring how commodity prices, macroeconomic variables, and policy decisions jointly evolve (Hamilton, 1983; Kilian, 2009; Baumeister and Peersman, 2013; Arezki et al., 2017; Alquist et al., 2020). We relate this to the renewed focus on external supply shocks in monetary economics (Bernanke et al., 1997; Blanchard and Gali, 2007; Bernanke and Blanchard, 2025), while showing how financialization reshapes these relationships (Tang and Xiong, 2012; Henderson et al., 2015; Basak and Pavlova, 2016). We then turn to the endogenous perspective, emphasizing asset-price channels and identification advances (Gertler and Karadi, 2015; Nakamura and Steinsson, 2018; Swanson, 2021) and the role of unconventional policies in influencing commodity markets (Anzuini et al., 2013; Frankel, 2014; Hammoudeh et al., 2015; Miranda-Pinto et al., 2025). We conclude with policy implications and directions for future research.

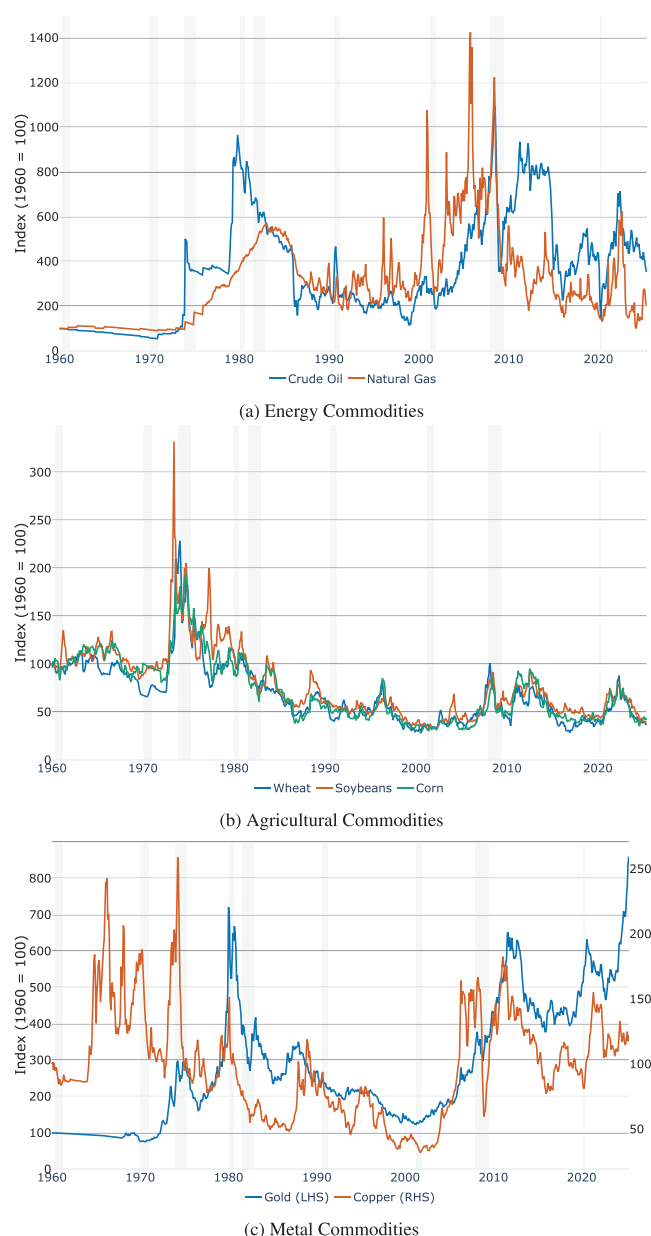
## 2 | Commodities and Monetary Policy

Commodities represent a potent source of macroeconomic fluctuations (Bogmans et al., 2024). In particular, commodity production typically relies on fixed natural resources, which constrain supply adjustments in response to short-term market fluctuations. Moreover, commodities' role as inputs in manufacturing, transportation, agriculture, and energy production represents another obvious source of macroeconomic impact. In what follows, we explore the three dimensions most relevant to central banks, namely the effects of commodities on inflation, their role in amplifying or dampening business cycle fluctuations, and the financial stability risks they pose.

### 2.1 | The Historical Evolution of Commodity Prices: Some Stylized Facts

As early as the pre-1913 era, Gerlach and Stuart (2024) show that globally integrated commodity markets already transmitted inflation across borders. Half a century later, the "Great Inflation" of the mid-1960s to early 1980s, and particularly the oil price shocks of 1973 (triggered by the Yom Kippur War and subsequent OPEC embargo) and 1979 (following the Iranian Revolution and subsequent Iran-Iraq War in 1980), brought greater scrutiny to the role of (energy) commodities in driving consumer price inflation and business-cycle dynamics (Hamilton, 2013). As shown in Figure 1, the 1973 shock led to a broader appreciation of commodity prices, generating similarly explosive increases in real agricultural commodity prices, gold, and high-grade copper. Although some scholars emphasized broader monetary or endogenous factors at the time as explanations for rising consumer prices (Barsky and Kilian, 2004; Bernanke et al., 1997), there is little doubt that major oil-market disruptions reverberated across other commodity markets and substantially influenced both inflation and the business cycle.<sup>1</sup> As illustrated by the shaded areas in Figure 1, each of the major energy shocks in 1973, 1979, and 1990 materialized only a few quarters before output contracted, indicating the predictive relevance of energy costs for recessions (Hamilton, 1983, 1985).

As influential as these commodity shocks were during the Great Inflation, their macroeconomic implications did not remain constant (Kilian, 2009; Kapetanios and Tzavalis, 2010; Baumeister and Peersman, 2013). From the mid-1980s to the early 2000s, in particular, the impact of commodity shocks was relatively more subdued (e.g., Blomberg and Harris, 1995; Hooker, 2002; Barsky and Kilian, 2004; Mishkin, 2007; de Gregorio et al., 2007). For instance, the 1990 oil price shock, triggered by the Persian Gulf War, had a relatively muted effect on both headline and core inflation in most advanced economies (Chen, 2009; Clark and Terry, 2010) despite a doubling of real oil prices in 5 months. This is also the era of the Great Moderation, characterized by factors such as reduced energy intensity and improved monetary policy frameworks and policy rules (Blanchard and Gali, 2007; Blinder and Rudd, 2013), which contributed to a period of macroeconomic stability.<sup>2</sup> Formal evidence mirrors this chronology. Bai-Perron structural break tests isolate sharp regime shifts in commodity prices, particularly gold, in 1971:01, 1980:12, and 2012:05, coincid-



**FIGURE 1** | Inflation-adjusted commodity prices (1960–2025).

[Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

This figure captures inflation-adjusted changes in selected commodity prices over the last six and a half decades. The constant-dollar price indices for energy, agricultural, and metal commodities are each rebased to 1960 = 100 and deflated with the US CPI from FRED. The series run from January 1960 through May 2025. The start date reflects the earliest month for which the World Bank Pink Sheet provides commodity prices. Shaded areas represent NBER recessions.

ing with the end of Bretton Woods, the Volcker disinflation, and the post-GFC safe-haven phase, respectively.<sup>3</sup>

After this period of relative calm, the mid-to-late 2000s brought renewed commodity-price volatility, as evidenced by the global food crisis of 2007/08 and subsequent shocks during the recovery from the GFC in 2011/12 (Headey and Fan, 2008). This episode is often characterized as the beginning of a new commodity supercycle (e.g., Cuddington and Jerrett, 2008; Carter et al., 2011; Erten and Ocampo, 2021). Prices for essential agricultural

commodities, including wheat, soybeans, and corn, more than doubled within months. Oil prices peaked at an all-time high of US \$147 per barrel in July 2008, shortly after US natural-gas prices reached a local maximum in June 2008, only months into the Great Recession, before collapsing precipitously toward the end of that year (Hamilton, 2009). Research attributes the 2007-08 surge chiefly to fundamentals, in particular, growing Asian demand, supply bottlenecks, and inventory dynamics, with continued debate about the auxiliary role of speculation, including index investing, as well as monetary conditions and regulatory frictions (e.g., Masters, 2008; Browne and Cronin, 2010; Irwin and Sanders, 2012; Hamilton and Wu, 2015; Brunetti et al., 2016).

Following the GFC, commodity return volatility eased through the mid-2010s, especially in energy. Yet, a sequence of large and persistent shocks in agricultural commodities during the late 2000s and early 2010s food supercycle underscored the macroeconomic importance of non-energy commodities. This shifting risk composition redirected research attention from an almost exclusive focus on oil toward the macroeconomic role of non-energy commodities (Cecchetti and Moessner, 2008; Browne and Cronin, 2010; Ferrucci et al., 2012; de Winne and Peersman, 2016; Conflitti and Luciani, 2019; Peersman, 2022). As illustrated in Figure 1, energy-price shocks have historically aligned closely with recessions, whereas agricultural and metal prices follow more frequent boom-bust patterns that only occasionally trigger broader macro-financial stress. For research, this argues for sector-specific, regime-dependent identification and forecasting rather than a single commodity proxy.

Recently, supply disruptions during the pandemic and geopolitical tensions following Russia's conflict with Ukraine have once again placed commodity markets at the forefront of macroeconomic debates (IMF, 2024; Baumeister, 2023). Subsequent rapid surges in crude oil, natural gas, and agricultural commodity prices in 2022 quickly fed into higher headline inflation, affected core inflation and expectations, and even precipitated secondary effects through upward pressures on wage demands (Bernanke and Blanchard, 2025). Cross-country evidence shows that in 21 economies, energy-price shocks alone accounted for the bulk of headline-inflation surprises in 2021-23, with core inflation rising chiefly through pass-through rather than excess demand (Dao et al., 2024). Metal markets exhibited distinct behavior, with gold prices setting new records since 2020, consistent with a safe-haven demand channel in response to macroeconomic uncertainty. Copper prices also rose by more than 50% in less than a year, pointing to industrial demand. The general surge in commodity prices following the pandemic has raised concerns about the onset of another commodity supercycle (Monge and Lazcano, 2022; Ginn, 2023).

Table 1 summarizes the main pass-through mechanisms of commodity shocks as well as the policy-relevant variables they influence, along with representative studies. We elaborate on each in the remainder of this section.

## 2.2 | Commodities and Inflation

Macroeconomic theory typically interprets commodity price shocks, particularly oil price shocks, as exogenous adverse supply shocks. This implies a direct cost-push channel, where increases in commodity prices feed into production costs, which firms pass on to consumer prices (e.g., Peersman and van Robays, 2009). Such dynamics may lead to persistently higher price levels. Moreover, if labor markets are sufficiently flexible, subsequent wage-push inflation may emerge as workers demand higher wages to offset the erosion of their real incomes. This again leads to rising costs for producers and, ultimately, higher consumer prices. In addition, several studies suggest that energy shocks can also act as adverse demand shocks, as higher energy costs squeeze disposable income and curb spending (Edelstein and Kilian, 2009; Baumeister and Kilian, 2016; Baumeister et al., 2018). Because energy is an indispensable commodity, households facing high marginal propensities to consume experience substantial real-income losses during energy price shocks, significantly reducing their purchases of other goods, depressing aggregate demand, and potentially inducing recessions even without monetary tightening (Auclert et al., 2023).

Generally, energy and food commodities have been found to be substantial drivers of inflationary pressures (Blomberg and Harris, 1995; Ferrucci et al., 2012; Peersman, 2022; Ginn, 2023). Their importance stems from the sizable share they constitute in consumer expenditures and their tendency to create spillover effects along the production chain, influencing the prices of a broader range of goods. Borrallo et al. (2024), for example, document that food and energy price shocks have substantial pass-through effects on European Union food inflation, with the impact peaking roughly 12 months after the initial shock. Diaz et al. (2024) similarly find that energy price shocks have become increasingly potent in advanced economies since the GFC, though Japan appears to be an exception. Castelnovo et al. (2024) explore the differential impacts of food and oil price shocks. They find that food price shocks chiefly generate second-round wage and core inflation effects (an adverse supply shock). In contrast, oil price shocks tend to escalate financial uncertainty and dampen durable and energy-related consumption (an adverse demand shock). Notably, the authors attribute the larger second-round impact of food shocks to a pro-cyclical monetary policy response that reinforces the initial impact. In contrast, oil shocks are often met with countercyclical measures that limit inflation.

Substantial empirical challenges remain in identifying the pass-through of commodity prices, as they are determined in competitive and flexible markets that adjust much faster than the consumer goods prices they influence (Ferrucci et al., 2012). As retail and industrial prices typically respond more slowly due to menu costs, contract rigidities, or market power, the timing and magnitude of pass-through effects have proved difficult to isolate. In addition, fiscal policy measures such as export bans, subsidies, and carbon-reduction strategies can also create or exacerbate commodity market shocks, with ripple effects on consumer prices (Kpodar and Liu, 2022). In this context, Ferrucci et al. (2012) emphasize the importance of accounting for country-specific policy frameworks, especially commodity price controls



**TABLE 1** | Commodities, macroeconomic outcomes and financial stability.

| <b>Mechanism</b>                          | <b>Policy-relevant variable(s)</b> | <b>Description</b>                                                                                                                                                                                                | <b>Representative Studies</b>                                                                                                                                                   |
|-------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cost-push pass-through                    | Inflation                          | Higher commodity prices raise firms' marginal costs, which are (partly) passed through to consumer prices.                                                                                                        | Peersman and van Robays (2009): SVAR with sign restrictions; Borralló et al. (2024): EU threshold-VARX; Peersman (2022): Euro Area SVAR-IV                                      |
| Input-output/Supply-chain amplification   | Inflation, Output                  | Energy and key agriculture inputs raise downstream costs, broadening and prolonging pass-through beyond direct fuel/food components.                                                                              | Kilian and Zhou (2023): narrative + sectoral channels; Alquist et al. (2020): survey of energy macro linkages                                                                   |
| Real-income squeeze (adverse demand)      | Inflation, Output                  | Costlier energy/food erode households' real income, curbing consumption and investment.                                                                                                                           | Auclert et al. (2023): HANK model; Edelstein and Kilian (2009): IRF via Cholesky-identified SVAR; Baumeister et al. (2018)                                                      |
| Wage-price effects                        | Inflation                          | Workers seek higher wages and firms may raise markups to offset real-income/cost shocks, embedding the shock in core inflation.                                                                                   | Bernanke and Blanchard (2025): four equations hybrid SVAR; Castelnuovo et al. (2024): SVAR with sign restrictions; Jain et al. (2024): survey evidence                          |
| Expectations                              | Inflation                          | Households extrapolate from frequently observed prices of everyday commodities (groceries, fuel, etc.); when those salient prices rise, expectations rise too, thereby adding persistence to aggregate inflation. | D'Acunto et al. (2021): OLS with microdata; Jo and Klopach (2025): DiD of state-level gas-tax holidays; Beaudry et al. (2025): dynamic factor model                             |
| Exchange-rate pass-through                | Inflation, Output                  | Shifting exchange rates alter local-currency prices of dollar-invoiced commodities. Pass-through varies with import dependence and dominant-currency pricing.                                                     | Flaccadoro (2024): structural dynamic factor model; Ding et al. (2023): VAR and DCC-GARCH                                                                                       |
| Monetary-policy framework/Credibility     | Inflation, Output                  | Credible, well-anchored monetary regimes attenuate shock pass-through to inflation and output; weak credibility amplifies it.                                                                                     | Choi et al. (2018): local projections; López-Villavicencio and Pourroy (2019): dynamic panel GMM.                                                                               |
| Terms-of-trade/Dutch-disease reallocation | Employment/Output                  | Commodity booms appreciate the currency, shifting resources from tradables; busts reverse the pattern.                                                                                                            | Bodenstein et al. (2018): panel VAR; Branstetter and Laverde-Cubillos (2024): dynamic panel GMM; Mien and Goujon (2022).                                                        |
| Exporter credit/Sovereign-spread          | Output, Financial stability        | Commodity booms narrow sovereign spreads, easing credit; busts widen spreads, tightening credit, and amplifying economic and financial cycles.                                                                    | Drechsel and Tenreiro (2018): two-sector small-open-economy DSGE; Wegener et al. (2025): weekly predictive regressions.                                                         |
| Price-volatility/Uncertainty              | Output                             | Commodity-price volatility raises macro uncertainty and real-option values, delaying irreversible investment and hiring.                                                                                          | Bjørnland et al. (2018): medium-scale DSGE with nominal rigidities; van Eyden et al. (2019): dynamic panel growth model                                                         |
| Bank balance sheet                        | Financial stability                | Adverse price shocks weaken commodity-exposed balance sheets, raising NPLs and default risk.                                                                                                                      | Kinda et al. (2018): panel fixed-effects regressions; Eberhardt and Presbitero (2021): GARCH + Mundlak-logit panel model; Agarwal et al. (2020): fixed-effects panel regression |

(Continues)

TABLE 1 | (Continued)

| Mechanism                              | Policy-relevant variable(s)    | Description                                                                                                                                | Representative Studies                                                                                                                                                                 |
|----------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Leverage & margin-call spirals         | Financial stability            | Sharp price swings trigger margin calls, forcing deleveraging and fire-sales that lead to liquidity spirals cascading through markets.     | Financial Stability Board (2023): Case studies & descriptive margin-flow/network analysis.; Daskalaki and Skiadopoulos (2016): event study; Fernandez-Perez et al. (2023): event study |
| Financialization & contagion           | Financial stability            | Tightening co-movements within commodities and across commodity–financial markets transmit price shocks broadly, amplifying systemic risk. | Nazlioglu et al. (2015): GARCH + VAR-based IRF; Diebold et al. (2017): spillover index; Apostolakis et al. (2021): Dynamic connectedness via TVP-VAR                                   |
| Climate/Geopolitical supply-shift risk | Inflation, Financial stability | Climate events or fragmentation raise supply-shock frequency, adding inflation and credit stress.                                          | Flori et al. (2021): VAR; Baumeister (2023): Bayesian SVAR                                                                                                                             |

*Note.* This table summarizes the principal mechanisms through which commodity price shocks influence inflation, output, and financial-stability outcomes, together with representative empirical studies.

Abbreviations: DCC = dynamic conditional correlation; DiD = difference-in-differences; DSGE = dynamic stochastic general equilibrium model; GARCH = generalized autoregressive conditional heteroskedasticity; GMM = generalized method of moments; HANK = heterogeneous-agent New-Keynesian model; IRF = impulse-response function; LP = local projections; Mundlak-logit = panel logit model with Mundlak adjustment. NPL = non-performing loans; SVAR = structural vector autoregression; SVAR-IV = SVAR with instrumental variables; threshold-VARX = VARX with regime thresholds; TVP = time-varying parameters; VAR = vector autoregression; VARX = VAR with exogenous variables;

and taxes, when measuring the macroeconomic consequences of global commodity fluctuations (see also Peersman, 2022).

Khalil (2022) demonstrates that oil price fluctuations frequently originate from manufacturing-sector shocks and global trade linkages, highlighting the sectoral drivers behind commodity shocks. While commodity shocks caused by supply disruptions, such as geopolitical conflicts or natural disasters, may be short-lived, they can generate lasting macroeconomic effects, as evidenced by the oil price shocks of the 1970s. By contrast, commodity price fluctuations caused by aggregate demand shocks, such as those arising from changes in consumer behavior or policy, tend to exert more transient macroeconomic effects (e.g., Siklos, 2021). There is an ongoing debate over the relevance of supply and demand shocks in causing commodity price fluctuations, as well as the resulting effects on macroeconomic outcomes.

### 2.2.1 | Varieties of Pass-Through Effects

Despite a broad consensus on the positive link between commodity prices and inflation (e.g., Ciner, 2011; Gospodinov and Ng, 2013; Feldkircher and Siklos, 2019; Iddrisu and Alagidede, 2020; Peersman, 2022; Borralló et al., 2024), considerable differences emerge in the estimates of the speed, persistence, and overall scale of pass-through effects from commodities to inflation. The extent of pass-through varies systematically with country-specific fiscal policies, market structures, labor market institutions, and evolving global energy interdependencies.

In many emerging market economies, where food constitutes a large portion of household expenditure, food price increases account for a larger share of aggregate inflationary pressure

(Pourroy et al., 2016; Catão and Chang, 2015; Ginn and Pourroy, 2019). Energy-intensive economies may exhibit similarly pronounced sensitivities. Fasanya and Awodimila (2020) demonstrate this for Nigeria, a large oil producer, where a Phillips curve model augmented with energy prices forecasts inflation more accurately than one incorporating non-energy commodity prices. By contrast, for South Africa, incorporating non-energy commodity prices yields better predictive power, which the authors attribute to the country's relatively lower energy intensity (see also Gelos and Ustyugova, 2017). Similarly, Kpodar and Liu (2022) demonstrate that gasoline price shocks have a smaller but more persistent impact on inflation in advanced economies compared to developing countries, with repercussions for household purchasing power (see also Gelman et al., 2023). Especially in lower-income settings, food and energy expenditure shares are higher, making price shocks in these categories particularly regressive and prone to triggering social and political tensions. A recent example of this is the 2007/08 global food price crisis, which sparked protests in over 30 countries and contributed to the social and economic pressures that fueled the Arab Spring.

The inflationary effects of commodity price shocks also depend on an economy's trade position in a given commodity (Filis and Chatziantoniou, 2013). Salisu et al. (2017), for example, observe more substantial oil price-driven inflation in oil-importing countries than in oil-exporting ones. This asymmetry is attributable to the heightened pass-through of international oil price shocks into domestic cost structures in import-dependent economies, in contrast to the revenue-augmenting effects that mitigate inflationary pressures in exporting countries.

Beyond these direct pass-through differences, commodity price swings also alter a country's terms of trade, and hence its

exchange rate, which in turn affects domestic price levels. Commodity exporters' currencies tend to appreciate when their key export prices rise, and depreciate when those prices fall (Akram, 2009; Zhang and Baek, 2022). Such fluctuations may feed back into domestic inflation, as currency depreciation raises the local currency cost of imported goods (Aron et al., 2014). Empirically, however, the strength of this channel varies markedly across economies. Živkov et al. (2019) find only marginal effects of oil prices on inflation in Central and Eastern Europe. This more muted response may reflect the region's diversified energy mix and macroeconomic stabilization policies, dampening the transmission of external oil price fluctuations to domestic inflation. In another regional context, Zakaria et al. (2021) show that oil shocks have a significant positive and persistent influence on inflation in South Asia. These results underscore the pronounced vulnerability of South Asian economies to external commodity price disturbances driven by high import dependency and limited energy alternatives.

Even in a closed economy, the nature of pass-through can evolve over time. Bańbura et al. (2023) document that gas price shocks had a much more substantial impact on Euro area core inflation during COVID-19 than earlier, highlighting that not all energy shocks are alike (see also Alessandri and Gazzani, 2025). Whereas oil and other energy prices once exhibited extensive, rapid, and symmetric pass-throughs (Meyler, 2009; Baumeister and Kilian, 2016), divergences between them (see Figure 1a) imply that policymakers should not assume a uniform energy market shock driven solely or even primarily by oil. Peersman et al. (2021) also demonstrate that spillovers among oil, food, and other commodities have intensified over time, further complicating the assessment of commodity price pass-through. The bottom line is that interactions between various energy and non-energy commodities appear to be increasingly complex, making the distinct study of their macroeconomic implications imperative (Kilian and Vigfusson, 2011).

Heterogeneity at the sectoral and industry levels also shapes the pass-through of commodity prices to consumer prices. Recent evidence by Afrouzi et al. (2024) corroborates this, highlighting the role of sectoral differences in input shares and price-setting rigidity in governing the persistence and asymmetry of inflationary responses. Where profit margins absorb cost increases, the pass-through to final consumer prices is muted even though producer prices have increased (Jiménez-Rodríguez and Morales-Zumaquero, 2022). The extent of this absorption depends on market structure and competition (Afrouzi et al., 2024), leading to possible asymmetries not only in changes to the relative level of prices but also concerning positive and negative commodity price shocks. Price increases (so-called "rockets") are argued to pass through more quickly and fully than price declines ("feathers") (Borenstein et al., 1997; Kilian and Vigfusson, 2011; Ferrucci et al., 2012; Choi et al., 2018; Li and Guo, 2021; Borralló et al., 2024).<sup>4</sup> Large or prolonged commodity price surges thus reach consumers more readily, while more minor or short-lived fluctuations may be absorbed or offset (Ball and Mankiw, 1994; Balke et al., 1998; Kinnucan and Forker, 1987). In addition, key inputs may be bought on fixed-term contracts while prices are elevated, meaning that costs may remain higher even after the original shock has subsided (Bailey, 2023).

Labor market conditions add another layer of complexity (Bernanke and Blanchard, 2025; Gelos and Ustyugova, 2017). In flexible labor markets, rising commodity prices can rapidly spark higher wage demands, reinforcing the initial cost-push pressures. More rigid labor markets, by contrast, may moderate pass-through to inflation but risk higher unemployment (see Section 2.2). Accordingly, cross-country variability in wage-setting institutions has produced divergent inflation outcomes in response to global commodity shocks (Peersman, 2022). Identifying consistent empirical patterns, however, remains challenging. While the research mentioned above confirms that labor market structures shape the amplification or dampening of commodity price shocks, others find no robust evidence of such effects (Gelós and Ustyugova, 2017). Moreover, evidence suggests that while workers adjust wage expectations in response to higher inflation, they generally do not anticipate a one-for-one pass-through, thereby tempering second-round effects unless the labor market is unusually tight (Jain et al., 2024).

### 2.2.2 | Inflation Expectations and the Role of Monetary Policy

An increasingly prominent strand of research highlights the critical role of inflation expectations in amplifying or dampening commodity price shocks. Even brief spikes in the cost of key commodities, especially energy and food, can lead to persistent inflation if households and firms anticipate that higher prices will persist. Such changes in expectations alone suffice to shift wage demands, firm pricing strategies, and public perception, thereby setting in motion a self-reinforcing mechanism of persistently rising prices (Coibion and Gorodnichenko, 2015; Narayan et al., 2017; Orlowski, 2017; Binder and Rodrigue, 2018; Coibion et al., 2020; Känzig, 2021). Importantly, Beaudry et al. (2025) demonstrate that households react primarily to the breadth of price increases they experience. This dynamic has been evident during the post-pandemic period (Bernanke and Blanchard, 2025), where shocks to non-core prices initially boosted headline inflation, which eventually spread to a broad spectrum of durable goods and services (Kilian and Zhou, 2023).

Research particularly highlights the importance of non-core commodities for inflation expectations due to their effects on experienced inflation. Frankel (2006) argues that while the concept of "core inflation" is not widely understood by the general public, they focus extensively on salient price shifts such as those at gasoline stations or grocery stores. This is confirmed by D'Acunto et al. (2021), who use scanner data merged with a bespoke expectations survey to demonstrate that the frequency-weighted inflation each household actually pays for its grocery basket is a far stronger predictor of 12-month-ahead inflation expectations than the standard expenditure-weighted (core) CPI. In particular, the substantial effects of grocery price surprises on inflation expectations appear consistent with an attention channel. Evidence from US gas-tax holidays further corroborates this attention channel, showing that a 1% drop in gasoline prices lowered 1-year-ahead inflation expectations by roughly 0.13 percentage points, with larger declines when the price cut was explicitly publicized (Jo and Klopach, 2025).

In addition to their direct impact on inflation and inflation expectations, non-core price movements also exert a range of

indirect influences, as non-core goods underpin the production processes of virtually all other goods and services. As Kilian and Zhou (2023) argue, commodity-related inflationary impulses can emerge from multiple segments of the energy sector alone, further amplifying downstream cost pressures in the manufacturing of plastics and chemicals, as well as raising transport costs (see also Alquist et al., 2020). In the case of oil, for example, price changes affect not only transportation fuels, such as gasoline, diesel, and jet fuel, but also the costs of lubricants, asphalt, and an array of petrochemicals used in manufacturing plastics, pharmaceuticals, and synthetic textiles. Such channels are paralleled in agricultural commodities, which are central not only to food production but also to countless downstream sectors, including animal feed, biofuels, paper and packaging, textiles, adhesives, and even cosmetics, thereby amplifying inflationary pressures across multiple supply chains.

Greater central bank credibility has generally altered the pass-through from commodity prices to (expected) consumer inflation (Blanchard and Riggi, 2013; Blinder and Rudd, 2013). Choi et al. (2018), for instance, document a decline in the impact of oil price shocks on consumer prices from the 1970s to 2015, attributing this in part to better-anchored inflation expectations.<sup>5</sup>

A longstanding debate centers on whether central banks should respond to commodity price shocks. Classic arguments (Bernanke et al., 1997; Bernanke and Gertler, 2001; Cecchetti and Moessner, 2008) suggest that monetary authorities should generally focus on persistent shocks that threaten to de-anchor medium-term inflation, rather than chasing every short-lived price movement. Sudden policy tightening after a transitory surge, for example, can depress output and employment, creating an unnecessarily volatile macroeconomic environment. However, the events following the pandemic have reinforced that even transitory price shocks can become sticky if they unanchor inflation expectations (Bernanke and Blanchard, 2025). Once wage- and price-setters anticipate prolonged inflation, self-reinforcing second-round effects may materialize, even if the initial commodity shock recedes. Thus, while the challenge for central banks lies partly in distinguishing between transitory shocks and more permanent shifts that risk de-anchoring inflation expectations, it, more importantly, lies in preventing second-round effects from materializing (Cecchetti and Moessner, 2008). Nonetheless, complexities persist, particularly in emerging market economies prone to higher baseline inflation and exchange rate volatility (Anand et al., 2015; Furceri et al., 2016). An incorrect assessment of the persistence of a commodity shock in these economies can further undermine the central bank's credibility and necessitate more forceful tightening later (Cecchetti and Moessner, 2008).

The evidence highlights that the effectiveness of monetary policy in addressing commodity-induced inflation depends critically on the shock's underlying drivers, prevailing policies, and the anchoring of inflation expectations. In addition, commodity-specific shocks can differ extensively in their macroeconomic effects (Baumeister and Kilian, 2016; Aastveit et al., 2023; Ginn, 2023; Beaudry et al., 2025). More generally, when real-time data are noisy, the robust design of policy rules can be at least as consequential as responsiveness (Benchimol, 2024). In the face of this, central bank credibility and clear communication are cheap and essential stabilizers in containing second-round inflationary

effects by anchoring inflation expectations firmly. Specifically, when households and firms trust that persistent deviations from inflation targets will not be tolerated, their wage and price-setting behaviors remain moderated, mitigating second-round inflationary pressures.

## 2.3 | Commodities and the Business Cycle

Beyond influencing consumer prices, a large body of empirical work has studied how commodity price movements, particularly oil prices, are linked to economic fluctuations. Hamilton (1983, 1985)'s seminal studies demonstrate that oil price increases preceded most US recessions after World War II. This observation triggered an extensive literature exploring both the mechanisms through which energy price shocks contribute to economic contractions and the evolving nature of these effects over time (e.g., Bernanke, 1983; Bruno and Sach, 1982; Gisser and Goodwin, 1986; Lee et al., 1995; Hooker, 1996; Hamilton, 2003, 2009; Kilian and Vigfusson, 2011).

Although earlier work often emphasized oil's singular importance, subsequent research has extended the focus to a broader range of commodities and highlighted diverging macroeconomic repercussions for advanced and emerging economies (Fernández et al., 2018; Baumeister and Kilian, 2016). Such studies generally agree on at least two stylized facts. First, commodity price surges act as adverse demand shocks. They can hamper growth by curtailing real income, consumption, and investment, particularly when commodities figure prominently in households' budgets or firms' production costs (Bernanke et al., 1997; Hamilton, 2009; de Winne and Peersman, 2016; Herrera et al., 2019). Second, while substantial, the macroeconomic impact of commodity price shocks is highly context-dependent, varying across countries and over time (Fernández et al., 2017; Castelnuevo et al., 2024; Di Pace et al., 2025). For example, while Hamilton (2009) contends that the run-up in oil prices in 2007/08 helped shape the US GFC recession, others argue that oil has played a greatly diminished role in recession dynamics since the mid-1980s Great Moderation, partly owing to better-anchored inflation expectations and more flexible labor markets (Blanchard and Gali, 2007; Nakov and Pescatori, 2010; Peersman and van Robays, 2012).

The empirical evidence shows that the commodity-business cycle linkages are typically stronger in emerging economies, given their narrower export bases and greater dependence on commodity revenues (Kose, 2002; Blattman et al., 2007; Drechsel and Tenreiro, 2018). For example, Fernández et al. (2018) show that commodity price shocks in Brazil, Chile, Colombia, and Peru in the post-2000 period explain a large share of aggregate fluctuations, ranging from output growth to the real exchange rate and trade balance. Given its prominent but evolving role in commodity markets, China occupies an important place in these analyses. However, extant research reaches divergent conclusions on the magnitude and persistence of China's role (see, e.g., Tang et al., 2010; Klotz et al., 2014; Wei and Guo, 2016; Cross and Nguyen, 2017).

### 2.3.1 | Exchange Rates, Dutch Disease, and Financial Amplification

Unexpected exchange-rate movements mediate how commodity-price shocks transmit to the real economy. For net importers,



an unexpected rise in world commodity prices worsens the terms of trade by raising import prices relative to export prices. Under a floating rate, the higher import bill typically increases demand for foreign currency and puts downward pressure on the domestic currency, producing a nominal depreciation and, absent offsetting relative inflation, a real depreciation. In the absence of an offsetting monetary response, this real depreciation reallocates resources toward exportables and import-competing industries while simultaneously raising import-price inflation via exchange-rate pass-through. However, as de Gregorio et al. (2007) note, policymakers face a clear trade-off, since a freely floating exchange rate acts as a shock absorber but directly transmits volatility into domestic prices, whereas an exchange-rate smoothing regime tempers headline inflation at the expense of depleting reserves and accumulating external imbalances. Bobasu et al. (2025) show that the exchange-rate response to a shock hinges on where the central bank sets the real policy rate relative to its neutral level. When the bank keeps the real rate broadly unchanged, the exchange rate is free to move and absorb the shock. By contrast, a forceful hike in the real rate pushes the currency higher, deepens the downturn, and exacerbates inequality.

Among commodity exporters, by contrast, unexpected commodity price surges act as a wealth injection, lifting national income, real wages, and investment. Yet they almost inevitably push up the real exchange rate, squeezing the competitiveness of non-resource tradables, described as the classic Dutch-disease mechanism first formalized by Corden and Neary (1982) and reaffirmed in later cross-country evidence (Sachs and Warner, 1995; Badeeb et al., 2017; Mien and Goujon, 2022). While the model predicts a contraction of the tradable sector and a rise in unemployment, empirical evidence is mixed. During the 2002–2008 commodity super-cycle, for instance, unemployment actually fell in Australia and Canada even though one can observe a clear contraction in the tradeable sector, underscoring the cushioning roles of domestic demand and policy institutions (Banks, 2011; Carney, 2012; Komarek, 2016).<sup>6</sup> Panel-VAR evidence by Bodenstein et al. (2018) shows that positive commodity-price shocks can even tighten labor markets in advanced small open economies by drawing resources toward booming sectors via an appreciating real exchange rate.

Turning to firm- and region-level data, the effects are markedly heterogeneous. Firm-level data from Colombia, for instance, reveal that boom-driven appreciations persistently suppress manufacturing R&D and technology upgrading (Branstetter and Laverde-Cubillos, 2024). Evidence from Indonesia indicates that whether manufacturing is crowded out or crowded in hinges on the labor intensity of extraction. While capital-intensive mines may boost local industry, labor-intensive mines are often displaced (Pelzl and Poelhekke, 2021). Broader sectoral-reallocation studies suggest that flexible labor markets, evolving consumption patterns, and active policy can offset, or mask, the employment losses traditionally associated with Dutch disease (Unel and Upton, 2023; Kulish et al., 2024). That being said, distributional effects remain pronounced. Commodity downturns depress real earnings for less-skilled workers in resource-rich regions (Benguria et al., 2024) and rising energy and food prices disproportionately burden low-income households (Gnocato, 2025).

For commodity-exporting economies, there is, in addition, growing evidence of a financial amplification channel. When export prices rise, investors lower the perceived sovereign-default risk for commodity-rich economies, compressing spreads and making foreign credit cheaper (Fernández et al., 2018; Dominguez-Cardoza et al., 2022). Easier external financing then feeds directly into domestic investment and consumption, amplifying the real boom (Drechsel and Tenreyro, 2018; Kohn et al., 2018). Quantitative work shows that this borrowing-cost channel explains a sizable share of GDP, consumption, and investment fluctuations in small open exporters (Drechsel and Tenreyro, 2018; Juvenal and Petrella, 2024; Fernández et al., 2018). The mechanism runs in reverse when commodity prices slump, as sovereign and corporate spreads widen, risk premia rise, and cross-border credit dries up, deepening the downturn (Fernández et al., 2018; Dominguez-Cardoza et al., 2022; Alessandri and Gazzani, 2025). Because upswings and downswings are both magnified through country-specific borrowing constraints, commodity exporters face stabilization challenges that are often sharper than those confronting net importers. Moreover, the spread-commodity nexus can flip signs when a price surge coincides with elevated geopolitical risk, so higher oil prices may tighten rather than ease credit conditions (Wegener et al., 2025).

Beyond price level shocks, commodity price volatility has attracted attention as a crucial determinant of macroeconomic uncertainty and stability. Classic real-options and growth research shows that greater volatility depresses investment and lowers long-run growth by raising the value of waiting (Pindyck, 1991; Ramey and Ramey, 1995; Blattman et al., 2007; Elder and Serletis, 2010). When investment is irreversible, this channel is reinforced, because higher uncertainty delays capital spending and output (Bernanke, 1983; Cavalcanti, Tiago V. de V. et al., 2015; Wang et al., 2022). Empirical research finds that spikes in oil-price volatility routinely precede or coincide with US recessions (Bjørnland et al., 2018), while a 144-year OECD panel shows that oil-price volatility cuts GDP growth, especially in exporting economies (van Eyden et al., 2019). In general, these effects are nonlinear and asymmetric, meaning that adverse volatility shocks hurt output more than favorable shocks help it (Federer, 1996). Comparable patterns hold for other primary goods as well, and the growth penalty is largest for low-income countries reliant on a narrow export base (Blattman et al., 2007). In addition, firm-level evidence shows that information that raises perceived macro-uncertainty leads managers to cut hiring, investment, and production plans (Kumar et al., 2023). Moreover, research by Diercks et al. (2024) shows that closely spaced uncertainty shocks interact in a super-additive way, so a cluster of shocks inflicts far larger losses than an isolated occurrence. Taken together, volatility amplifies the contractionary force of adverse commodity shocks while muting the stimulus from positive ones, raising formidable stabilization challenges for policymakers.

### 2.3.2 | The Role of Monetary Policy

Commodity price shocks create acute policy trade-offs for inflation-targeting central banks. Under a strict headline inflation targeting (IT) rule, an adverse commodity supply shock that lifts CPI while depressing activity would, for example, still trigger tightening, risking a procyclical response. Such a scenario is illustrated by the ECB's hikes in 2008 and again in April/July

2011 and is consistent with US evidence that the Fed historically tightened after food-supply shocks (but eased after oil shocks) as core inflation and expectations rose (de Winne and Peersman, 2016).

Given these trade-offs, optimal monetary policy favors a flexible, state-contingent approach over a rigid, one-size-fits-all rule. In Taylor-rule terms, this means allowing the weights on inflation and the output gap to vary with the type and persistence of the shock (see also the value of robust rule design under limited information in Benchimol, 2024). For example, a supply-driven commodity price spike that raises inflation but hurts growth may warrant tolerating first-round effects (or focusing on core inflation) to avoid a deep output slump, while guarding against a deanchoring of inflation expectations. By contrast, a demand-driven commodity boom, lifting both inflation and activity, may justify a stronger tightening bias to preempt overheating. For commodity exporters, explicitly incorporating terms-of-trade movements is often beneficial. After a price crash, the currency typically depreciates, and pass-through raises headline inflation even as income falls, so a strict headline IT rule could mechanically tighten. An open-economy-aware rule, by contrast, would instead ease to cushion activity (Ferrero and Seneca, 2018). More generally, terms-of-trade shocks are amplified under pegs, whereas flexible regimes better buffer growth (Bleaney and Greenaway, 2001; Edwards and Levy Yeyati, 2005). In sum, policy should be context-aware, distinguishing supply versus demand drivers, transient versus persistent swings, and importer versus exporter situations, to balance price stability with output stabilization.

## 2.4 | Commodities and Financial Stability

Commodity investments have historically offered distinct advantages, including a low correlation with equities and fixed-income securities, the potential for higher returns, and a positive correlation with inflation (e.g., Gorton and Rouwenhorst, 2006; Salisu et al., 2019). These attributes attracted investors pursuing portfolio diversification, consequently driving the financialization of commodity markets and transforming commodities from production inputs into prominent financial assets (Belousova and Dorfleitner, 2012; Cheng and Xiong, 2014; Adams and Glück, 2015; Chari and Christiano, 2019; van Huellen, 2019; Adams et al., 2020).

Over time, however, these benefits have diminished. Specifically, the correlation between commodities and conventional financial assets has risen sharply over the past two decades (Silvennoinen and Thorp, 2013; Büyüksahin and Robe, 2014; Adams and Glück, 2015; Maghyreh et al., 2016; Adekoya and Oliyide, 2021; Mensi et al., 2022; Alam et al., 2022; Liu et al., 2023). Moreover, interdependence among commodities has more than tripled in the aftermath of the GFC, remaining persistently elevated thereafter (Kang et al., 2017; Diebold et al., 2017; Zhang and Broadstock, 2020; Delle Chiaie et al., 2021; Balcilar et al., 2022).

While the financialization of commodity markets has enhanced market liquidity and depth, it has concurrently intensified the risk of financial contagion and the susceptibility of the global financial system to systemic disruptions caused by commodity shocks. The remainder of this section reviews the role of balance sheet stress, leverage-driven liquidity spirals, and contagion, as

well as the sharpened trade-offs between the previously discussed objectives and financial stability.

### 2.4.1 | Balance Sheets, Liquidity, and Contagion

Several channels link commodity market shocks to financial instability. First, severe volatility or declining prices negatively impact the balance sheets of commodity-exposed firms, impairing debt-servicing capacity and thereby deteriorating banks' asset quality (Kinda et al., 2018; Eberhardt and Presbitero, 2021). The high concentration among key financial intermediaries, such as global banks engaged in clearing and financing, magnifies the systemic repercussions of individual failures. Second, extensive leverage in commodity trading can generate liquidity spirals as margin calls compel traders to exit positions or seek additional financing precisely when credit conditions tighten (Financial Stability Board, 2023). Empirical evidence further confirms that margin hikes reduce price momentum, drain market liquidity, and spill over into related commodity contracts, exacerbating liquidity stress during volatile episodes (Daskalaki and Skiadopoulos, 2016). Third, the opacity of over-the-counter derivatives markets complicates regulators' and market participants' efforts to quantify and monitor institutional exposures (Financial Stability Board, 2023).

Some of these vulnerabilities became evident during the 2020–2022 period, characterized by pandemic-related supply-chain disruptions and rising geopolitical tensions. These factors culminated in sudden spikes in commodity prices, widening spreads among different grades of the same commodity, margin calls, as well as liquidity pressures that mutually reinforced each other (Financial Stability Board, 2023; IMF, 2023). The May 2020 WTI negative-price episode,<sup>7</sup> for instance, vividly illustrates how extreme margin calls and depleted storage capacity trigger forced deleveraging and liquidity spirals (Fernandez-Perez et al., 2023). Importantly, Fernandez-Perez et al. (2023) find that passive commodity index investors were not responsible for the negative-price crash, challenging narratives that link financialization-related passive index investing directly to extreme price instability. Although recent episodes largely avoided widespread financial contagion, they underscore the opaque and concentrated nature of commodity trading, clearing, and financing and the risk this can pose for financial stability.

The link between commodity shocks and systemic financial stress has been examined in a small but growing literature. For instance, Nazlioglu et al. (2015) identify bi-directional volatility spillovers between oil prices and financial stress before and after the GFC, indicating that commodity shocks exacerbate systemic risk during periods of macroeconomic instability. Similarly, Apostolakis et al. (2021) document how oil price volatility permeates the financial system, generating macro- and micro-level uncertainties, including increased financing costs for firms and shifts in investors' risk preferences. Liu et al. (2021) find the impact of oil price shocks on China's financial stress to be contingent upon the prevailing volatility regime. Additionally, commodity price shocks profoundly impact banks' balance sheets, especially in emerging and developing economies reliant on commodity exports (Agarwal et al., 2020). Adverse commodity shocks rapidly elevate non-performing loans, undermining bank profitability and increasing financial distress (Kinda et al., 2018).

Agarwal et al. (2020) further show that adverse commodity price shocks disproportionately constrain credit supply among banks with low deposit bases and higher nonperforming loans.

Beyond these short-term liquidity and credit channels, commodity shocks can also fuel wider macro-financial volatility through a “financial accelerator” mechanism (Bernanke et al., 1999). As uncertainty rises or terms of trade worsen, default risk can escalate, tightening credit conditions further and triggering flights to safer assets (Brown and Yücel, 2002). The broader implication is that abrupt commodity price movements can set off self-reinforcing loops of currency pressure, asset repricing, and financial stress. Nonetheless, Kilian and Zhou (2022) caution that the US dollar remains relatively insulated from global oil price changes, suggesting that commodity-induced exchange rate fluctuations may have a more subdued impact in the US compared to many emerging or smaller developed economies that heavily depend on the commodity trade.

Finally, climate-related risks heighten financial stability concerns. As climate change alters production and consumption patterns in agricultural and energy commodities, more frequent and severe price swings may translate into additional strains on financial institutions and markets (Flori et al., 2021). Research on the direct link between climate-induced commodity shocks and financial stability is still nascent. However, its salience likely rises as climate and transition risks amplify commodity volatility. Beyond climate risks, the IMF (2023) has cautioned that the ongoing fragmentation in physical commodity markets could further heighten financial market volatility and trigger abrupt exchange rate adjustments.

As central banks increasingly integrate financial stability objectives alongside inflation and employment targets, incorporating commodities into policy frameworks becomes imperative. However, key questions remain unanswered. For instance, the role of leverage in amplifying systemic risk, the precise contagion channels between commodities and other asset classes, and the efficacy of macroprudential and monetary policy tools in containing commodity-related financial instability all deserve further exploration.

#### 2.4.2 | The Role of Monetary Policy

Central banks in advanced economies, for which liquidity tools are readily available and credibility is high, increasingly separate their objectives by pairing policy rate adjustments with targeted stabilization policies. During the March 2023 US banking turmoil, for example, the Fed maintained its tightening path but offered emergency liquidity in the form of the Bank Term Funding Program to supply 1-year collateralized loans and stem deposit outflows. Likewise, the Bank of England’s temporary gilt-market purchase operations in 2022 stabilized long-dated UK government bonds even as policy rate hikes continued. Moreover, faced with commodity-related margin shocks, the Bank of England went a step further, unveiling the £40 billion Energy Markets Financing Scheme in October 2022 to backstop energy firms’ collateral needs without derailing its tightening cycle. Such instrument segregation, however, is far more difficult for small, open, commodity-dependent economies, where acute exchange-rate pressure, fiscal fragility, and lower policy credi-

bility often force them into an “all-in” interest-rate defense or pro-cyclical tightening that risks amplifying commodity shock (Bouis et al., 2025). Exporter-importer asymmetries also extend to policy reaction functions. In oil busts, exporters tend to tighten while importers ease, and pro-cyclical responses become more aggressive when global financial uncertainty is high (Yildirim and Guloglu, 2024). Hence, monetary policy challenges regarding links between commodities and inflation outside the advanced group of economies may well be greater.

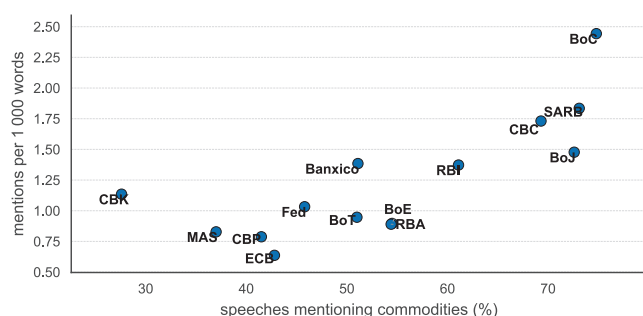
Nevertheless, the policy frontier lies in pre-emptive, multi-layered buffers to protect against (commodity-induced) financial instability. Counter-cyclical capital and liquidity tools, such as the Basel III counter-cyclical capital buffer and sector-specific margin add-ons, raise loss-absorbing capacity in booms and can be released in busts to sustain credit, complementing the “lean-against-the-wind” stance when asset-price and credit cycles overheat (Kockerols and Kok, 2021). System-wide liquidity and climate stress tests, now routinely run by the Bank for International Settlements, the ECB, and other members of the Central Banks and Supervisors Network for Greening the Financial System (NGFS), integrate commodity and climate scenarios into banks’ balance-sheet assessments (Baudino et al., 2024). Moreover, the ECB’s forthcoming “climate-factor” collateral framework will hard-wire such considerations into daily lending operations. Beyond monetary policy, fiscal and external buffers matter just as much. Sovereign wealth and stabilization funds, for example, can smooth public spending and foreign reserve positions over the price cycle, dampening pro-cyclical fiscal swings in commodity exporters. Ultimately, while the inflationary and business-cycle implications of commodity price shocks are now well articulated in the literature, the financial stability channels, including the impact on bank capital, credit spreads, liquidity spirals, and contagion, remain understudied, highlighting a pressing need for further research to inform policy design.

### 3 | Commodities in Central Bank Communication

In this section, we examine how central banks incorporate commodity markets into their official communications. This analysis illuminates policymakers’ perceptions, traces how the discussion has evolved across institutions and over time, and allows us to compare approaches and assess their alignment with the academic literature. For this analysis, we draw on a range of recent speeches by central bankers contained in the CBS dataset (Campiglio et al., 2025).<sup>8</sup> More details on the analysis can be found in Appendix B, which provides further empirical evidence and identifies the senior officials and central banks cited in this section.

Beginning with a quantitative analysis of the speeches, Figure 2 plots each central bank’s share of speeches that contain at least one commodity reference (“breadth”) against the average number of references per 1000 words (“depth”). Three patterns emerge. First, central banks of commodity-exporting countries, such as the Bank of Canada, the South African Reserve Bank, and the Central Bank of Chile (upper-right quadrant), talk about commodities both frequently and at greater length, underscoring the macro-critical role of terms-of-trade shocks in their policy narratives. Second, major advanced economies, such as the ECB and the Fed, tend to mention commodities episodically and with





**FIGURE 2** | Commodity discourse across central banks.  
[Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/joes.12005)]  
The figure draws on the CBSpeeches corpus of Campiglio et al. (2025). We analyze 14 central banks: Bank of Canada (BoC), South African Reserve Bank (SARB), Reserve Bank of India (RBI), Reserve Bank of Australia (RBA), Board of Governors of the Federal Reserve (Fed), Bank of England (BoE), European Central Bank (ECB), Bank of Japan (BoJ), Central Bank of the Philippines (CBP), Monetary Authority of Singapore (MAS), Bank of Thailand (BoT), Central Bank of Kenya (CBK), Central Bank of Chile (CBC) and Bank of Mexico (Banxico). The filtered sample contains 13,643 speeches (1986–2023). Commodity language is identified by counting case-insensitive matches of the tokens: commodity/ies, raw material, oil, crude oil, natural gas, gasoline, energy, gold, silver, copper, iron ore, steel, metal(s), wheat, corn, soybean(s), rice, food price, agriculture. The horizontal axis measures the percentage of each authority's speeches that mention at least one commodity term. The vertical axis measures the average commodity mentions per 1000 words.

modest depth, while the Monetary Authority of Singapore lies at the low end. Third, central banks of large commodity-importing economies, which experience a pronounced pass-through of global commodity prices into domestic cost structures, notably the Reserve Bank of India and the Bank of Japan, frequently refer to commodities in their speeches.

Broader diagnostics confirm this taxonomy. Speech-length distributions reveal that central bank speeches of commodity-exporting countries are shorter and consequently achieve higher mention density, whereas the Fed, ECB, and Bank of England issue much longer texts in which commodity language is episodic. Moreover, when commodities matter for macro policy, they are discussed by governors, accounting for about half of mentions in Canada and South Africa, whereas in committee-led or diversified regimes, the topic is delegated to deputies or board members. Interestingly, at monthly frequency, contemporaneous and lead-lag ( $\pm 3$ -month) correlations between speech density and commodity returns are small, suggesting calendar and threshold effects dominate.

Qualitatively, we analyze speeches of the past two decades that frequently reference commodities. Throughout these speeches, commodity price fluctuations are primarily characterized as exogenous cost shocks. Recently, policymakers have cited global supply disruptions, geopolitical risks, and shifts in global demand, highlighting events like the COVID-19 pandemic and the escalating Russia-Ukraine conflict, to illustrate that energy and food price movements lie beyond domestic policy control (Kuroda, 2022; Cunliffe, 2022; Gravelle, 2022; Bailey, 2023). For instance, Bernanke (2008) attributed pre-GFC high oil prices mainly to global demand surpassing supply. Similarly, Kuroda (2022,) noted that for net energy importers, “prices of

commodities such as crude oil and natural gas are determined by supply and demand in global markets; they cannot be controlled by the Bank's monetary policy” (p. 6). Consequently, policymakers emphasize that monetary responses must consider global conditions and local supply-chain information, ultimately aiming to mitigate the second-round effects of commodity shocks (e.g., Khan, 2015; Bailey, 2023).

While each central bank addresses commodities in relation to its economy's structural characteristics, policymakers generally discuss commodity markets in terms of (headline) inflation and, secondarily, the real economy and financial stability (e.g., George, 2020; Lane, 2022; Haskel, 2022; Bailey, 2023; Pill, 2023). Commodity-driven price spikes, especially in energy and food, are noted for disproportionately impacting lower-income households, who allocate a larger share of their disposable income to essentials. Cunliffe (2022), for instance, highlights that energy expenses for the bottom 10% of households are already more than twice the average share, making these groups particularly vulnerable to reductions in consumption during energy-price shocks.

Another frequently discussed channel is the terms-of-trade effect. Policymakers from commodity-importing countries, such as the UK (Cunliffe, 2022; Pill, 2023), Japan (Kuroda, 2022), and the euro area (Lane, 2022), characterize rising global energy costs as an external tax that transfers domestic income abroad. This loss in terms of trade reduces domestic purchasing power and puts pressure on headline inflation. Policymakers in these economies emphasize flexible exchange rates as a means of shock absorption (Cunliffe, 2022; Lane, 2022). Kuroda (2022) further argues that energy-price shocks from the Russia-Ukraine conflict justify continued easing rather than tightening, given subdued domestic demand and limited second-round wage-price effects in Japan (see also Wakatabe, 2022). Conversely, policymakers in commodity-exporting countries like Canada (Carney, 2008; Schembri, 2016; Gravelle, 2022) note that high commodity prices improve terms of trade and positively influence growth. Nevertheless, they stress careful management of commodity booms to prevent excessive currency appreciation, macroeconomic imbalances, and unwanted sectoral shifts (Vergara, 2014; Patterson, 2016).

A common refrain among policymakers is that central banks cannot and should not offset every commodity price shock (Yellen, 2011; Kuroda, 2022; Lane, 2023; Cunliffe, 2022). Lane (2023), for example, notes that energy and food price shocks tend to be large but transient. Therefore, monetary policy should prioritize persistent inflation components, carefully monitor pass-through effects, and avoid excessive tightening (Dhingra, 2023). Policymakers frequently emphasize that interest-rate policy has limited influence on supply-driven commodity shocks. Even for demand-driven shocks, monetary policy acts with a lag, meaning that by the time demand is curbed, the shock may have already subsided. Consequently, central bankers have, since the early 2000s, preferred core inflation metrics as a more accurate gauge of underlying pressures in the face of commodity shocks (George, 2021; Waller, 2023).

That being said, policymakers caution that if inflation expectations risk becoming unanchored, central banks must respond decisively. Tenreiro (2022) and Haskel (2023) note



that while ignoring transient energy-price spikes can prevent unnecessary tightening, persistent shocks may embed inflation more deeply. Therefore, monetary policy should adjust gradually, targeting medium-term CPI stability. (Pill, 2023, p. 5) similarly warns that repeated transitory shocks can accumulate “through time into a more long-lasting elevation of headline inflation.” Policymakers identify persistent second-round effects, such as rising wage demands and markups, as primary threats (Tenreiro, 2022; Pill, 2023; Dhingra, 2023). In addition, Schnabel (2020) highlights an opposite challenge, where low energy prices, though beneficial in reducing inflation and boosting output, can prolong the ECB’s effort to re-anchor inflation expectations at a higher level, necessitating clear communication of a lengthened “medium-term” policy horizon.

Some policymakers explicitly highlight that commodity shocks, particularly those in the energy sector, can pose a threat to financial stability. Cunliffe (2022), for instance, extensively discusses increased margin calls on commodity derivatives and the “chilling effect” (p. 6) of financial sanctions linked to Russia’s invasion of Ukraine. Extreme commodity-price volatility in particular is thought to trigger unexpected liquidity demands on producers, traders, and financial intermediaries, potentially affecting banks and clearinghouses if margin frameworks are insufficient (Bowman, 2019; Schulhofer-Wohl, 2023). Tarullo (2014) further notes that commodities may expose the banking sector to legal, environmental, and price risks through banks’ commodity-related activities. Groepe (2015) similarly identifies commodities as a cross-border stability risk in Africa. Policymakers like Bostic (2022) and Panetta (2023) emphasize that central banks and regulators must closely monitor liquidity pressures in commodity financing, maintain smooth functioning of the derivatives market, and prevent excessive risk transfer to banks and other entities.

Another recurring theme is that climate change and policy responses will reshape commodity markets. Policymakers frequently link increased commodity-price volatility to extreme weather events and the transition away from fossil fuels. Schnabel (2022), for example, describes commodity prices as the primary channel through which climate-transition policies and geopolitical developments are transmitted into euro-area inflation. She argues that intensified climate action implies persistently elevated fossil fuel prices, placing upward pressure on headline inflation that central banks cannot simply “look through.” Instead, policy normalization must carefully support the green transition and ensure financial markets accurately price carbon risks. Similarly, Panetta (2022) and Mann (2023) emphasize that shifting from fossil fuels to renewables could generate new types of supply shocks. They highlight short-term price surges in critical metals such as lithium, cobalt, copper, and rare-earth elements, driven by rapid growth in clean-energy technologies. Boer et al. (2024) also estimate that net-zero emissions scenarios could lead to sustained price increases for these metals, potentially complicating the energy transition and increasing their strategic significance to that of oil. Menon (2022) similarly identifies commodity markets as central to climate challenges, with fossil fuels as the problem and critical minerals as new constraints. Central banks thus face the task of adapting their monitoring frameworks, interpreting commodity-price signals more precisely, steering capital toward greener investments, and cushioning the inflationary and distributional impacts of the green transition.

Mostly absent from these discussions is sustained consideration of how monetary policy itself might influence commodity market dynamics. Central banks typically characterize commodity-price fluctuations as exogenous shocks (Lane, 2015; ECB, 2025). The few exceptions that acknowledge indirect central bank influences are illustrative. Tarullo (2014), for example, highlights how monetary policy might indirectly shape banks’ lending into commodity infrastructure projects, generating systemic risks. Panetta (2022) similarly notes that monetary policy can impact innovation and investment choices, thereby affecting the dynamics of fossil-fuel supply and demand. Schnabel (2022) describes “greenflation” as an endogenous demand shock driven partly by monetary policy, as private investment responds to climate action. Menon (2022) treats carbon pricing explicitly as a policy lever, though underlying fossil-fuel costs remain external. Haskel (2022) briefly acknowledges the potential endogenous amplification of commodity shocks on domestic inflation and growth, while Jain (2019) emphasizes the role of domestic policy in shaping commodity market outcomes through credit access and technology adoption, rather than external shocks. Nevertheless, central banks predominantly constrain themselves to addressing commodity price effects after the fact, rather than considering a proactive influence on commodity dynamics. This framing implies that policymakers monitor but do not perceive their policies as materially affecting commodity prices.

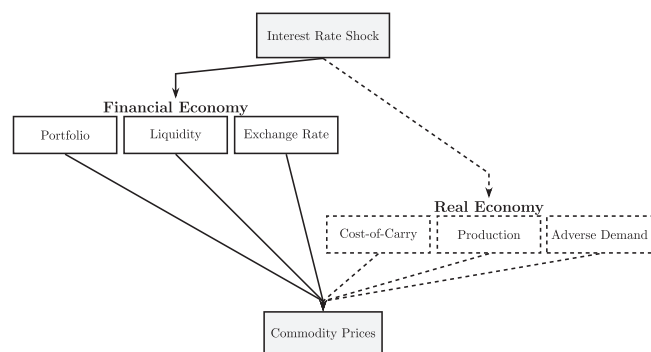
## 4 | Commodities in the Monetary Transmission Mechanism

Treating commodity prices as exogenous to monetary policy overlooks a key two-way interaction. Specifically, monetary policy shocks move commodity prices as much as they move other asset prices, and the financialization of commodity markets has tightened these links as energy, metals, and grains increasingly trade like standard asset classes. It is well known that easy monetary conditions have at times coincided with commodity booms and greater speculative activity (Filardo et al., 2020). Yet the literature on monetary policy surprises has largely centered on equities, bonds, and exchange rates rather than commodities (Gürkaynak et al., 2005; Nakamura and Steinsson, 2018; Altavilla et al., 2019; Swanson, 2021). This omission is striking because commodities play a dual role, both as investable financial assets and as production inputs, which sets them apart from the markets mentioned above. As a result, monetary policy can transmit not only through financial channels (portfolio rebalancing, discount rates and funding costs, and exchange rates) but also uniquely through real economy channels (inventory carry costs and convenience yields, extraction and investment timing, and final demand). Such a transmission has implications for forecasting commodity prices, the design of optimal monetary policies, and also for the international transmission of monetary policy via commodity markets (Juvenal and Petrella, 2024; Degasperis et al., 2025; Ider et al., 2024).

### 4.1 | Transmission Channels

#### 4.1.1 | Interest Rates

Although stylized and abstracting from informational frictions, the canonical view suggests an inverse relationship between interest rates and commodity prices. In particular, interest rate



**FIGURE 3** | Interest rate shock transmission to commodity prices.

This figure illustrates how an unexpected change in interest rates propagates to commodity prices through two clusters of mechanisms. Financial economy channels, portfolio rebalancing, liquidity and risk-taking, and the exchange-rate channel, react quickly and are shown on the left. Real-economy channels, inventory cost-of-carry, production/extraction timing, and the adverse demand channel, unfold more slowly and appear on the right. Solid arrows denote the financial cluster, dashed arrows the real cluster. All paths ultimately converge to commodity prices, underscoring that both sets of channels jointly determine the net price response to an interest rate shock, although at different frequencies.

shocks, arising from unexpected changes in the policy rate or longer-term rates influenced by unconventional monetary policies, affect commodity prices through both financial and real economic channels. Figure 3 provides a stylized summary of the transmission channels frequently highlighted in the literature.

The financial economy channel, where agents continuously trade and arbitrage across asset classes, predominantly drives high-frequency responses in commodity prices. Conversely, the real economy channel adjusts more gradually, influencing commodity prices through inventory holding costs, timing decisions in resource extraction and production, and ultimately, the interest-sensitive components of final demand. Although forward-looking agents theoretically revise production, inventory, and consumption plans immediately upon a policy surprise (Miranda-Pinto et al., 2025), the real-side adjustment is slowed by fixed production schedules, investment delays, contractual and technological rigidities that impede rapid price transmission, as well as imperfect foresight regarding the shock's ultimate economic impact. Consequently, commodity prices are likely to reflect only a partial real-economy adjustment at the moment of policy announcement.

**Portfolio Channel.** Beginning with the financial economy channels, the portfolio channel posits that rate shocks alter the relative returns on commodities vis-à-vis other assets. In response, investors shift their holdings away from assets whose relative expected return has fallen (e.g., Tobin, 1969; Mishkin, 1995; Yip et al., 2024). In particular, when a policy shock raises the risk-free rate, the relative expected return on commodity investment falls, while the financing cost of margin-funded futures rises (Ivan et al., 2025). Faced with imperfect substitution across asset classes, asset managers and leveraged traders scale back long commodity exposures and shift toward relatively higher-yielding bonds and cash equivalents, such that the prices of commodities decline almost immediately. To add complexity, the magnitude of

these declines depends on (i) which segment of the yield curve moves, (ii) how the shock interacts with funding constraints and margin calls, and (iii) whether the commodities concerned serve as inflation- or geopolitical-hedge assets. This portfolio-rebalancing effect has strengthened with the financialization of commodity markets. Greater use of derivatives, commodity index funds, and margin financing has amplified speculative participation and synchronized orders, magnifying small rate shocks into sizable price changes.

**Liquidity Channel.** The liquidity or risk-taking channel complements the broader portfolio-rebalancing mechanism by emphasizing that monetary easing works not only through relative expected returns but also by relaxing funding constraints and expanding investors' appetite for risk (e.g., Belke et al., 2013; Zhu et al., 2018; Adams et al., 2020; Bianchi et al., 2020; Apergis et al., 2020; Bauer et al., 2023). Lower policy rates or large-scale asset purchases boost market liquidity, compress margins and value-at-risk limits, and induce traders and asset managers to enlarge positions in high-beta assets, including commodity futures. Conversely, rate hikes tighten funding conditions, trigger margin calls, and force deleveraging, putting immediate downward pressure on futures prices.

**Exchange Rate Channel.** Because nearly all traded raw materials are invoiced in US dollars, any shock that shifts US interest-rate differentials typically also moves the US dollar and, through that exchange-rate change, commodity prices. In particular, a stronger dollar raises the effective foreign-currency cost of commodities, thereby curbing global demand. At the same time, the appreciation boosts local-currency revenues for commodity exporters and can elicit additional supply. As the dollar moves, arbitrage aligns local-currency prices with the new dollar benchmark, propagating the US shock globally (Miranda-Pinto et al., 2025). The net effect is therefore an inverse relationship between the dollar and dollar-denominated commodity prices, well-documented for both conventional and unconventional policy episodes (Schuh, 1974; Frankel, 1986; Rosa, 2014; Zhang et al., 2016; Glick and Leduc, 2018; Flaccadoro, 2024). Recent research, however, shows that the strength (and sometimes even the sign) of this exchange-rate channel depends on (i) dominant-currency pricing, which mutes pass-through into trade volumes (Adler et al., 2020) and (ii) structural shifts such as the United States' new status as a net energy exporter (Rees, 2025).

**Cost-of-Carry Channel.** Commodity markets occupy a hybrid position between financial and goods markets, and, as such, the literature has long emphasized the real channels through which monetary policy transmits. Frankel (1986), by extending Dornbusch (1976) overshooting model to commodity markets, was the first to show that even in a closed economy, an interest-rate shock alters the relative return on storable commodities, thereby moving their prices and highlighting storage as the pivotal margin of adjustment (see also Frankel, 2006). Unlike equities or fixed-income securities, commodities derive their spot value simultaneously from current use and from their asset value as claims on future scarcity. Commodity storage thus links today's supply-demand balance to tomorrow's, making inventories the key state variable through which monetary policy can affect spot and near-dated futures prices in ways that have no analogue in equity or bond markets (Fama and French, 1987).

Holding a unit of a storable commodity from today to a future date involves (i) physical carrying costs (warehousing, insurance, shrinkage); (ii) the opportunity cost of capital, which is proxied by the short-term risk-free rate, and is partly offset by (iii) the convenience yield, the non-pecuniary benefit of having the good on hand. The theory of storage<sup>9</sup> (Working, 1949; Brennan, 1976) predicts that in equilibrium, a rise in the risk-free rate widens the futures-spot spread and compresses desired inventories and pushes spot prices down (see also Glick and Leduc, 2012; Miranda-Pinto et al., 2025). Deaton and Laroque (1992, 1996) embed this cost-of-carry logic in a rational-expectations competitive-storage model, which links expected spot-price growth to short-term interest rates (see also Kleppe and Oglend, 2017). The model features a “stock-out” region, in which, once inventories hit zero, storage arbitrage breaks down and prices become dominated by contemporaneous supply shocks, such that rate shocks lose traction. The obvious assumption here is that the commodity considered can be stored, whereas perishable commodities or electricity should remain unaffected by this channel.

**Production Channel.** Complementing the cost-of-carry channel as a demand-side channel, the production/extraction-timing channel translates monetary shocks into supply-side responses. The underlying mechanism highlights that shifts in the real interest rate alter the timing of extraction and investment decisions in resource industries (Frankel, 2014; Anderson et al., 2018; Casassus et al., 2018). When the discount rate rises, the present value of leaving a barrel of oil or a ton of copper in the ground falls. Where operational constraints allow, producers increase their output. Conversely, a lower real rate encourages deferral of extraction, tighter current supply, and higher prices. Because capacity additions are lumpy and irreversible, lower real rates can bring forward investment, creating a scarcity-to-abundance regime shift that bends spot-price paths and convenience yields (Casassus et al., 2018). While this short-run response operates through the classic Hotelling arbitrage condition, longer-horizon capacity and exploration decisions are governed by irreversible investment under uncertainty and by the entire term structure of real interest rates.

**Adverse Demand Channel.** Finally, monetary tightening reduces aggregate demand by raising real borrowing costs, curbing credit growth, and undermining household and business confidence. Because commodities such as oil, metals, and certain agricultural staples are integral to the production of durables and intermediate goods, any downturn in spending feeds directly into diminished physical demand (Miranda-Pinto et al., 2025). Futures markets capitalize on this anticipated shortfall, adjusting forward prices promptly based on updated macroeconomic outlooks, often before actual consumption declines materialize.

#### 4.1.2 | Expectations

While the interest rate channel outlines a clear inverse relationship between interest rates and commodity prices, empirical observations (see Section 4.2) paint a much more nuanced picture. To understand these reactions, it is crucial to recognize that monetary policy goes beyond the mechanical setting of interest rates, since announcements also carry information about

the central bank’s assessment of the economy, thereby shaping market expectations (e.g., Blinder, 2018; Coenen et al., 2017; Bernanke, 2020). Commodity prices, being forward-looking, can hence move in the same direction as interest rates if markets react to the information revealed by policy. For example, an interest rate hike may be interpreted as “stronger demand,” lifting industrial metals prices. An interest rate cut, by contrast, may instead be understood as a “weak outlook,” depressing commodity prices despite easier financing.

Empirically, the challenge is to separate pure rate effects from information effects. High-frequency event studies provide a starting point by isolating monetary policy surprises via market reactions around central bank announcements (Kuttner, 2001; Gürkaynak et al., 2005; Gertler and Karadi, 2015; Altavilla et al., 2019; Swanson, 2021, 2023). However, the fundamental identification problem remains, because monetary policy surprises are a composite of policy action and information revelation. For instance, positive surprises can coincide with equity gains (Nakamura and Steinsson, 2018), reflect systematic responses to public data (Bauer et al., 2023), or capture shifts in risk sentiment (Kroencke et al., 2021). Reactions have also been found to be state-dependent and larger when pre-announcement uncertainty is high (Benchimol et al., 2023).

Identification has advanced along two lines. First, by augmenting high-frequency surprises with proxies for the central bank’s information set, including forecast revisions and private-sector nowcasts (Romer and Romer, 2000; Barakchian and Crowe, 2013; Gertler and Karadi, 2015; Miranda-Agrippino and Ricco, 2021; Acosta, 2023; Benchimol et al., 2023). Second, by exploiting joint asset responses in the announcement window to extract orthogonal factors, namely a “pure policy” component (yields up, equities down) and a “central-bank information” component (yields and equities up together) (Jarociński and Karadi, 2020). Broader factor approaches that include bonds, equities, FX, and commodity futures reveal a non-yield factor driving cross-market moves that purely yield-based surprises miss (Cieslak and Schrimpf, 2019; Kroencke et al., 2021; Boehm and Kroner, 2024).

Table 2 provides a summary of the salient transmission channels, including three expectations-driven channels most relevant to commodity markets. The analysis of such expectations-based channels in the context of commodity markets is scarce. A comparative analysis of expectations-driven modifiers across central banks in the context of commodity markets does not, to the best of our knowledge, exist.

## 4.2 | Empirical Evidence

Our synthesis of the empirical literature below proceeds in three steps. First, we benchmark evidence for conventional interest rate shocks in the dynamics of both aggregated price indices and disaggregated commodity prices. Next, we extend the analysis to unconventional policy. Thereafter, we focus on financialization, liquidity, and volatility channels, reviewing how policy-induced shifts in risk-taking and market structure modify price, correlation, and microstructure behavior. A selective summary of the most relevant studies can be found in Table 3.

**TABLE 2** | Monetary-policy transmission channels to commodity prices

| Channel                                     | Source | Locus      | Tightening | Speed     | Illustrative refs.                                          |
|---------------------------------------------|--------|------------|------------|-----------|-------------------------------------------------------------|
| <i>Financial-economy</i>                    |        |            |            |           |                                                             |
| Portfolio rebalancing                       | IR     | Fin.       | ↓          | min–hrs   | Tobin (1969); Ivan et al. (2025)                            |
| Liquidity/Risk-taking<br>(funding & margin) | IR     | Fin.       | ↓          | min–hrs   | Belke et al. (2013); Bianchi et al. (2020)                  |
| Exchange-rate pass-through                  | IR     | Fin.→ Real | ↓*         | hrs–days  | Rosa (2014); Flaccadoro (2024)                              |
| <i>Real-economy</i>                         |        |            |            |           |                                                             |
| Cost-of-carry (inventory)                   | IR     | Real       | ↓†         | days–qtrs | Frankel (1986); Deaton and Laroque (1996)                   |
| Production/Extraction                       | IR     | Real       | ↓          | qtrs–yrs  | Hotelling (1931); Anderson et al. (2018)                    |
| Adverse demand                              | IR     | Real       | ↓          | days–qtrs | Miranda-Pinto et al. (2025)                                 |
| <i>Expectations-driven modifiers</i>        |        |            |            |           |                                                             |
| CB information effect                       | I      | Exp.       | amb.       | min–hrs   | Nakamura and Steinsson (2018); Jarociński and Karadi (2020) |
| Store-of-value demand                       | IR+I   | Exp.       | amb.‡      | hrs–days  | Batten et al. (2014); Gomis-Porqueras et al. (2023)         |
| Risk-shift/Sentiment                        | I      | Fin./Exp.  | amb.       | min–hrs   | Boehm and Kroner (2024)                                     |

Note. Legend: IR = interest-rate component; I = information component; Fin. = financial-market mechanism; Real = real-economy; Exp. = expectations.

\* Signs refer to USD-denominated commodity prices. Local-currency import prices can move in opposite directions. Pass-through varies with dominant-currency pricing and structure.

† Mainly for storable commodities and weak near stock-outs. Perishable goods/electricity are largely unaffected.

‡ Under pure policy tightening (higher real rates, stronger USD), safe-haven metals tend to fall. Under Information/risk-off news they may rise.

**Conventional Monetary Policy.** Early vector autoregressive analyses with broad indices confirmed the inverse interest rate-price link, yet already hinted at heterogeneity. Using monthly vector autoregressions, Anzuini et al. (2013) find that a one-percentage-point easing lifts the aggregate CRB index by about 5% within half a year, with oil and metals reacting fastest. Hammoudeh et al. (2015) use a long quarterly sample to analyze sector-level responses, demonstrating that the average response remains negative after six quarters, while food prices persistently remain positive, highlighting the limitations of index evidence. Using disaggregated data, Scrimgeour (2015) shows that 17 spot contracts respond within minutes to FOMC surprises, yet with opposite signs across sectors. Miranda-Pinto et al. (2025) use Jarociński–Karadi surprises as external instruments and estimate that a 10 bp US rate hike cuts storable energy and base-metal prices by up to 5% in 3 weeks. They subsequently show that commodity prices transmit more than half of the ensuing CPI response. Comparable evidence for China suggests that quantity-based instruments matter more than the administered policy rate. Hammoudeh et al. (2024), for instance, document that an M2 expansion produces larger and longer-lived increases in non-fuel and metals indices than a symmetric rate cut.

Single-commodity studies refine these results. Ider et al. (2024) employ high-frequency surprises around 211 ECB meetings and show that a 10 bp tightening cuts Brent crude by about 3%. Counterfactuals in their proxy-SVAR prove that headline euro-area inflation would fall much less without the oil leg. Yu et al. (2023) complement this price evidence with regime-switching GARCH estimates, demonstrating that monetary-policy uncertainty itself

raises the probability of transitioning into a high-volatility regime for WTI. Taken together, conventional policy shocks matter most for storable industrial commodities and energy. Food, beverages, and some agricultural raw materials often deviate, indicating that real supply and inventory incentives interact with the cost-of-carry channel in commodity-specific ways (Belke et al., 2013; Alam and Gilbert, 2017; Yao and Alexiou, 2022).

**Unconventional Monetary Policy.** Glick and Leduc (2012) exploit 10 Federal Reserve and 5 Bank of England LSAP announcements and find that, contrary to popular belief, commodities fall on easing news even as long yields and the dollar decline, a result they attribute to recession signalling. Apergis et al. (2020) compare pre- and post-2008 regimes in a two-stage EGARCH-GMM framework and show that large-scale asset purchase (LSAP) shocks have effects two to three times larger than conventional rate moves on eight metals and energy prices, mainly because they magnify economic-uncertainty premia. Papadamou and Sogiakas (2017) study the effects of unconventional policy announcements from multiple central banks on precious metals. They find that the Bank of Japan's policies tend to stabilize and lift precious metals prices. At the same time, ECB announcements have more destabilizing effects, illustrating how investors' interpretations of the same policy type can differ across jurisdictions. Zhu et al. (2018) further illustrate this heterogeneity, showing that while the Federal Reserve's quantitative easing announcements strongly influence gold prices, similar announcements from the ECB have weaker effects, and those from the Bank of Japan have a negligible impact. Gomis-Porqueras et al. (2023) show that



**TABLE 3** | Empirical evidence on monetary policy transmission to commodity markets.

| Author(s)                     | Mechanism                                    | Sample                                    | Identification                            | Principal result                                                                | Policy view                                           | Key limits                               |
|-------------------------------|----------------------------------------------|-------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------|
| Apergis et al. (2020)         | Rates vs. UMP; uncertainty/leverage          | US +EA, 1989–2016, 8 metals/energy (d)    | ARMA-EGARCH-X, GMM on uncertainty         | UMP cuts returns, lifts vol. $\gg$ conventional                                 | Bigger, riskier moves under UMP                       | Metals/energy only; ends 2016            |
| Glick and Leduc (2012)        | LSAP/QE announcements (signalling)           | US +UK, 2004–11, GSCI indices (d)         | 1-day event with intraday surprises       | LSAPs $\downarrow$ commodities despite lower yields/FX                          | QE need not inflate commodities; signalling dominates | Short window; indices mask sectors       |
| Hammoudeh et al. (2015)       | Rate vs. QE-liquidity shocks                 | US, 1957–2013, IMF agg. +sectors (q/m)    | 6-var SVAR (FFR or reserves)              | Tightening hits metals/fuel; food rises; QE small                               | Target sectors; prefer core to headline CPI           | Indices hide dispersion; recursive order |
| Anzuini et al. (2013)         | Rate shocks via carry/inventory              | US, 1970–2008, CRB spot (m)               | 5-var SVAR + narrative checks             | 100 bp easing $\uparrow$ index 4–6 %, oil 7 %                                   | Easy policy not sole driver of super-cycles           | Pre-ZLB; broad index                     |
| Gomis-Porqueras et al. (2023) | FFR / FG / LSAP split                        | US, 1991–2019, 6 core commodities (d)     | 30-min HF PCs, local projections          | FG $\uparrow$ gold; LSAP $\downarrow$ gold                                      | Commodity pattern reveals UMP type                    | Six assets; pre-COVID                    |
| Hammoudeh et al. (2024)       | PBoC rate vs. M2 shocks                      | CN, 1996–2021, IMF indices (q)            | Bayesian SVAR                             | Rate hike $\downarrow$ composite $\approx$ 2 %; M2 $\uparrow$ non-fuel          | M2 tool stronger; caution on money supply             | Quarterly data; no fiscal link           |
| Smales (2019)                 | FOMC path surprise $\rightarrow$ liquidity   | WTI, 2011–16, 1-min book                  | $\pm$ 10-min event, panel                 | Liquidity dries pre-release, normalises 9 min                                   | Time margins/halts around FOMC                        | One commodity; liquidity only            |
| Yang et al. (2023)            | Policy vs. information shocks (oil)          | WTI/Brent (d, intraday window), 1994–2020 | HF policy shocks in SVAR with info factor | Tightening (pure policy) $\downarrow$ oil; positive CB info $\uparrow$ oil      | Separate rate from info shocks to avoid sign bias     | Energy only; identification window short |
| Yu et al. (2023)              | Policy-uncertainty $\rightarrow$ vol. regime | WTI, 2000–20 (d)                          | TVTP-MS-EGARCH w/MUI                      | Higher MUI $\uparrow$ high-vol regime prob.                                     | Cut uncertainty to calm markets                       | One asset; US index                      |
| Miranda-Pinto et al. (2025)   | Fed/ECB HF surprises                         | 39 futures 1990–2020 + macro (d/m)        | JK HF + proxy-SVAR                        | 10 bp hike $\downarrow$ prices 1–5 %; drives $\approx$ 60 % of CPI pass-through | Commodity channel pivotal, cross-border               | Only rate shocks; ends 2020              |
| Ider et al. (2024)            | ECB surprise $\rightarrow$ oil/gas           | Brent/TTF HF 2002–21                      | OIS event + BPSVAR                        | 10 bp hike $\downarrow$ Brent 3 %; bulk of CPI drop                             | Energy channel vital; front-load hikes                | Oil/gas only                             |
| Ordu-Akkaya and Soytaş (2020) | QE $\rightarrow$ equity–commodity spillovers | S&P 500 + 25 GSCI 1990–2017 (d)           | ADCC-GARCH + DY                           | Post-2010: commodities net vol. receivers; QE amplifies                         | Watch cross-asset risk when altering QE               | Volatility focus; Fed only               |

*Note.* This table collates key empirical papers on how conventional and unconventional monetary policies affect commodity market dynamics, outlining each study's identification strategy, main findings, policy implications, and limitations.

Abbreviations: q/m/d = quarterly/monthly/daily data; ADCC = asymmetric DCC-GARCH; ARMA = autoregressive–moving-average; DY = Diebold–Yilmaz spillover index; EGARCH = exponential GARCH; FFR = federal funds rate; FG = forward guidance; GMM = generalized method of moments; HF = high-frequency; JK = Jarociński–Karadi HF identification; LP = local projections; LSAP = large-scale asset purchases; MS = Markov-switching; MUI = Monetary-policy Uncertainty Index; PCs = principal components; PSVAR = proxy-SVAR (BPSVAR = Bayesian PSVAR); SVAR = structural vector autoregression; TVP = time-varying parameters; UMP = unconventional monetary policy.

“asset-like” commodities (e.g., gold, silver) respond more aggressively than production or agricultural commodities to unconventional monetary policy. Their results indicate that LSAPs can even reduce the prices of traditional safe-haven assets despite lower interest rates.

High-frequency decompositions confirm that not all unconventional monetary policy surprises are alike. Rosa (2014) finds sharp but inconsistent responses to policy surprises in high-frequency (5-minute) energy price data. Similarly, Basistha and Kurov (2015) analyze 129 FOMC announcements from 1994 to 2008 and observe

significant negative initial responses in crude oil and gasoline futures, although these effects dissipate within days. Gomis-Porqueras et al. (2023) separate fed funds, forward guidance, and LSAP shocks. They find that guidance easings lift gold and silver, while LSAP easings depress them, again pointing to conflicting growth versus liquidity signals. Ordu-Akkaya and Soytaş (2020) trace a complementary portfolio-balance channel, in which Fed asset purchases increase volatility spillovers from equities to 23 of 25 GSCI commodities, showing how cheap liquidity draws index investors into futures markets. Yang et al. (2023), analyzing high-frequency policy shocks within a structural VAR framework, explicitly distinguish monetary policy shocks from information shocks, showing that tightening surprises depress oil prices. In contrast, positive central bank signals regarding the economic outlook elevate them. These results align with Glick and Leduc (2012) findings, emphasizing that overlooking informational effects may bias estimated policy impacts. Nevertheless, Yang et al. (2023) conclude that monetary policy shocks contribute to forecasting oil price movements but do not constitute the primary drivers (see also Anzuini et al., 2013).

**Financialization, Liquidity, and Volatility.** Smales (2019) documents that bid-ask spreads in WTI futures widen, depth collapses, and order-book slopes flatten in the 10 minutes straddling FOMC releases. The deterioration scales with the size of the surprise and reverses within minutes, a pattern relevant for margin management and circuit-breaker design. Monetary-policy uncertainty amplifies volatility (Yu et al., 2023) and dampens news absorption (Kurov and Stan, 2018; Gospodinov and Jamali, 2018). Importantly, Gospodinov and Jamali (2018) highlight that adverse monetary policy shocks, coupled with high uncertainty, particularly amplify the impact on commodity prices. Meanwhile, Kurov and Stan (2018) show that when monetary-policy uncertainty is high, asset-price responsiveness to macroeconomic announcements strengthens in Treasury and foreign-exchange markets but weakens in equities and crude oil, suggesting that, for commodities, policy expectations transmit chiefly via the dollar and discount-rate channels. Gu et al. (2023) and Hayo et al. (2012) demonstrate that transparent communication can partially offset these effects, yet this effect diminishes when global risk aversion is high. Zaremba et al. (2021) report that accommodative policy strengthens cross-commodity co-movement, echoing the volatility-spillover findings of Ordu-Akkaya and Soytaş (2020) and implying that diversification benefits weaken precisely when policymakers loosen conditions. Taken together, these results imply spillovers from minutes around announcements to days and, via common-factor co-movement, up to monthly horizons.

The stylized inverse relationship between interest rates and commodity prices survives in the data, but only as a first-order approximation. Once storage costs, demand expectations, exchange-rate movements, and risk premia are taken into account, both the sign and the magnitude of the response are commodity-, instrument-, and state-dependent. Conventional rate hikes depress storable industrial commodities most clearly, whereas unconventional tools operate through signalling, portfolio balance, and liquidity channels that can produce opposite effects. Policy uncertainty itself shifts volatility regimes. Three implications follow. First, heterogeneity matters. Relying solely on core inflation for near-term assessment can underweight categories that react fastest to monetary news, like energy and traded intermediates, and that

shape expectations. Monitoring commodity-sensitive indicators (futures curves, inventory spreads, and exchange rates) is therefore essential even when the medium-term target is headline inflation. Second, because forward guidance and balance-sheet actions can move safe-haven and industrial commodities in opposite directions, communication should make clear whether the intent is to signal the outlook or to alter term premia and liquidity. Third, because liquidity thins and spillovers intensify around policy events, coordination with market infrastructure on margin trading and the timing of maintenance windows may reduce avoidable amplification.

### 4.3 | Varieties of Central Bank Responses

The transmission of monetary policy to commodity prices depends on which central bank acts and its global footprint in trade and finance. Accordingly, the relative weights on the financial-economy, real-economy, and expectations-driven channels in Table 2 likely differ across jurisdictions. Comparative evidence remains thin, and a systematic cross-central-bank analysis is still lacking. Even so, this section offers a first-order approximation of such cross-central-bank differences, grounded in the limited evidence currently available.

US dollar invoicing, deep capital markets, and the Fed's outsized role in the global financial cycle give US monetary policy impulses immediate traction through the financial economy channels (Miranda-Agrippino and Rey, 2020; Degasperi et al., 2025). Most empirical evidence on the transmission of monetary policy to commodity markets to date therefore pertains to US monetary policy. High-frequency identification, for example, shows that a 10 bp rate hike by the Fed lowers broad commodity futures prices by roughly 1%–5% within hours, accounting for more than half of the subsequent decline in US headline inflation (Miranda-Pinto et al., 2025). Because the US dollar is the dominant invoicing currency, the same shock also propagates abroad, with approximately 60% of the Fed-induced fall in foreign CPI occurring through lower commodity import prices. In short, the Fed's leverage over commodities is rapid, mediated mainly by financial markets, and strongly global.

For the ECB, Bank of England, and, to a lesser extent, the Bank of Japan, the same channels appear to operate, but with muted amplitude. Neither the euro nor the pound is a dominant invoicing currency in the commodity trade, and their financial markets, although deep, are less central to global portfolios. Empirical work for the euro area, for example, finds that a comparable 10-basis-point ECB surprise reduces Brent crude prices by approximately 3% (Ider et al., 2024) and transmits to domestic inflation primarily through the energy component. Exchange-rate pass-through, therefore, remains operative, but its international reach appears narrower and more concentrated in energy than in metals and agriculture. For Japan, in particular, yield-curve control likely mutes the domestic interest rate channel, making the yen the principal margin of adjustment for commodity-price pass-through.

China's outsized role in the global demand for metals and energy, together with still-developing financial markets, shifts transmission toward real-economy channels. Moreover, capital controls, the prominence of quantity instruments, and the

occasional use of state inventories or administrative measures can dampen or delay standard financial-market channels, reinforcing the real-economy dominance of transmission via demand and inventories (Miranda-Agrippino et al., 2020). Quantitative estimates suggest that a 1% contractionary M2 shock by the PBoC, for example, depresses global industrial commodity prices by roughly 10% over four to six quarters (Ferriani and Gazzani, 2025). Additionally, about 70% of the spillover to advanced-economy inflation operates through this commodity-price route. Broad-money shocks also dominate policy-rate shocks in China, reflecting the coexistence of quantity and price instruments in its dual-target framework (Hammoudeh et al., 2024).

Direct evidence on commodity-exporting economies is scarce, so we offer a first-order approximation. For small open commodity exporters, dominant-currency invoicing and limited global demand shares imply that domestic policy shocks are unlikely to move US dollar commodity prices on impact, unlike the Fed, and to a lesser extent, the PBoC. The primary transmission is instead domestic. A surprise tightening appreciates the currency, lowers local-currency commodity prices and resource-sector revenues, and tightens domestic financial and operating conditions. Portfolio/liquidity spillovers to global futures should be weaker given shallower financial markets and a smaller footprint of domestic investors. However, local funding costs can affect hedging, inventories, and marketing margins. Real-side effects likely operate through extraction timing and investment. Higher real rates may bring forward output and delay capacity expansion, whereas lower real rates may defer extraction and accelerate investment. These could affect global prices only when the exporter is a marginal supplier in a thin market, inventories are low, or supply elasticities are small. Information effects are likely modest internationally unless policy communications convey news about global demand. Netting these channels, transmission for exporters is mainly via exchange rate and credit domestically, with episodic, commodity-specific, and state-dependent global price effects.

## 5 | Conclusions

In this paper, we provide a comprehensive survey of the bilateral relationship between monetary policy and commodity markets. Our findings reveal that commodity prices are no longer a peripheral feature that central banks can choose to “look through.” Instead, we show that they are both barometers and transmitters of macro-financial stress, shaping inflation, the business cycle, and financial stability, even as their own paths are shaped by the policies and credibility of central banks. Recognizing this reciprocity is essential in an era of geopolitical fragmentation, climate transition, and heightened financialization.

On the commodities-to-policy side, four points stand out. First, even in today’s diversified, service-heavy economies, shocks in energy and food prices remain the dominant external force driving headline inflation. Moreover, even beyond energy and food, movements across commodity classes affect output and employment chiefly through real-income effects and heightened uncertainty. Second, the size, speed, and persistence of those pass-throughs are highly state-contingent, varying with the nature of the shock (supply versus demand), the country’s exposure (net importer or exporter, energy intensity, labor-market

structure), and the policy regime (credibility of the inflation target, exchange-rate flexibility). Third, financialization has tied commodity derivatives more closely to equity, bond, and wholesale funding markets. As a result, price swings transmit not only cost pressure but also balance-sheet and margin-funding stress, adding a financial-stability dimension to the classic inflation-output trade-off. Fourth, while most central bankers treat commodity prices as externally determined, the evidence shows that their own credibility and communication can govern second-round effects. When expectations are well-anchored, “looking through” temporary spikes can be optimal. When they are not, however, policy inertia risks entrenching inflation and amplifying macro-financial volatility. In addition to these four main conclusions, two further observations bear emphasis. First, commodity price volatility carries independent and asymmetric costs for economic activity. Second, for commodity exporters, higher export prices improve fiscal and external positions, thus compressing sovereign spreads and lowering borrowing costs for the public and private sectors. Accordingly, this sovereign-spread channel can substantially amplify commodity-driven business cycles for these economies.

On the policy-to-commodities side, three findings follow. First, monetary policy shocks transmit to commodity prices via a hierarchy of mechanisms. Rapid financial channels, such as portfolio rebalancing, funding-liquidity conditions, and exchange rate pass-through, operate within minutes. By contrast, slower real economy channels, including inventory cost-of-carry, extraction timing, and aggregate demand channels, unfold over days to quarters. Moreover, expectations modify and affect both of these sets of channels. Second, the strength and sign of transmission depend on the policy instrument, the commodity, and the originating central bank. High-frequency evidence shows that conventional US rate hikes depress storable industrial commodities within hours and, in some estimates, explain a sizable share of the subsequent CPI adjustment at home and abroad. By contrast, People’s Bank of China money-growth shocks transmit with multi-quarter lags through real-demand channels, while ECB and Bank of England surprises generate narrower, currency-centered responses. Unconventional policies further complicate identification. For instance, forward guidance may buoy safe-haven metals even as large-scale asset purchases push them down, highlighting the need to disentangle “pure” interest rates from “information” effects. Third, financialization amplifies this transmission, increasing cross-market co-movement and margin-linked liquidity fragility precisely when central banks intervene. Together, the evidence recasts commodities as an integral, policy-sensitive component of the global monetary transmission mechanism.

For policymakers, the key insight is that commodities are best understood not as peripheral indicators but as core state variables in the monetary system and an active leg of policy transmission. The textbook “look-through temporary commodity shocks” rule only holds conditional on real-time classification of each episode by source (supply vs. demand), scope (which commodities, with attention to storability and inventory conditions), and persistence (transitory vs. lasting). This requires a standing monitoring architecture to allow for precise identification. Once classified, policy should treat supply-driven spikes that sap demand with patience, while demand-driven surges

or cases with rising second-round risks warrant earlier tightening. Importantly, credibility and clear communication lower the pass-through from salient prices to expectations, allowing for more selective responses. Because financialization can turn large price swings into liquidity risks, interest-rate policy must be paired with macroprudential instruments. These include countercyclical capital and liquidity buffers, targeted margin and collateral requirements, stronger clearing safeguards and transparency in OTC markets, and the timing of margin changes away from policy windows to reduce procyclicality. A core result of this survey is that transmission differs across commodities (storable vs. non-storable, industrial vs. safe-haven, energy intensity), economies (importers vs. exporters, exchange-rate regime, financial depth), and policy instruments (rates, guidance, balance-sheet policy). Consequently, one-size-fits-all approaches risk asymmetric welfare costs. Rules and communications should be explicit about which sectors and prices are being stabilized and over what horizon. Finally, because monetary spillovers transmit via commodities and have real cross-border effects, policymakers should pursue ex-ante coordination and deliberate sequencing across interest rates, foreign-exchange operations, macroprudential buffers, and sovereign-wealth or stabilization funds, especially in commodity-exporting economies.

Looking ahead, future research on three fronts seems most pressing. First, on the commodities-to-macro side, research should move beyond single-commodity, single-country designs toward network and explicitly state-dependent models that condition on inventories, labor-market tightness, and exchange-rate regimes. Also, research to quantify heterogeneity across storable vs. non-storable goods (e.g., electricity) and across households/sectors to capture distributional incidence is needed. Such research should embed fiscal and trade shields (tax holidays, export bans, and reserve releases) and the sovereign-spread/credit channel for exporters. In addition, research that measures how climate- and transition-driven supply risks, particularly in critical metals and rare earth elements, alter pass-through and volatility could prove fruitful. Second, on the policy-to-commodities side, future research should sharpen identification to separate interest rate, balance sheet, and information shocks and compare expectation-driven modifiers across central banks. Such research should also allow for time variation in responses across commodities. Moreover, this should be paired with disaggregated futures/options evidence to map how funding constraints, leverage, and margin trading transmit policy news at high frequency. Third, additional research on infrastructure and policy design is warranted. The unifying aim should be a framework that monitors commodity shocks in real time, tailors responses to the shock's source and structure, and recognizes that central banks both absorb and sometimes initiate such shocks. Most importantly, this revolves around building a real-time "commodity shock indicator" that fuses high-frequency multi-market data with forecasts as well as micro-data on margins and OTC exposures to study liquidity spirals and to calibrate counter-cyclical margin/collateral tools and stress tests.

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## Endnotes

<sup>1</sup>See Dupor (2025) for a discussion of the Great Inflation and subsequent policy reactions.

<sup>2</sup>A notable exception during these otherwise tranquil times was the abrupt spike in natural gas prices in December 2001, which occurred just prior to the onset of the early 2000s recession, especially in the US.

<sup>3</sup>See Appendix A. Breaks are dated using Bai-Perron tests (Bai and Perron, 1998) on 12-month log changes in real commodity prices for the sample shown in Figure 1, with specifications including the federal funds rate, the US dollar real exchange rate, and the NY Fed GSCPI. The results are robust to using a real FFR. Perron single-break tests similarly flag early-1970s shifts in oil and copper and mid-2000s breaks in grains. There is also some evidence, via cointegration tests, that commodity price movements are attracted to each other in the long run.

<sup>4</sup>This is known as the "rockets and feathers" effect, first introduced by Bacon (1991) to explain the asymmetric price adjustments in the UK retail gasoline market.

<sup>5</sup>Using experimental data, Dräger et al. (2024) show that central banks can limit the pass-through of inflation shocks to inflation expectations through clear and credible forward-looking communication.

<sup>6</sup>In Australia, the 2002–08 upswing was led by bulk commodities, including iron ore and coking coal. The Australian terms of trade hit historic highs, and mining investment rose to its highest recorded share of the economy, alongside a strong AUD and a squeeze on manufacturing, even as aggregate unemployment fell. For Canada, the boom was centered on energy, as oil's share of commodity production rose from about 18% to 46% over 15 years, the CAD appreciated with global commodity prices, and manufacturing's GDP share declined from roughly 18% to 11% by 2012 (Carney, 2012).

<sup>7</sup>On April 20, 2020, the front-month WTI futures contract (May 2020) settled at a price of US dollar -37.63 per barrel, the first negative price on record, because traders facing scarce storage at the nearly full Cushing hub paid counterparties to assume delivery.

<sup>8</sup>The CBS dataset contains 35,487 unique speeches delivered by representatives of 131 central banks from January 1986 (the earliest available speech) to December 2023; see <https://cbspeeches.com> for details. To our knowledge, no prior work systematically examines commodity-specific central bank communication. Accordingly, the evidence reported here is a first-pass, descriptive analysis of commodity references. A dedicated NLP study of commodity language could prove informative and allow sharper comparisons with the broader communication literature (cf. Benchimol et al., 2025).

<sup>9</sup>See also Fama and French (1987); Jacks and Stuermer (2020); Fernandez (2020); Han et al. (2025) for additional discussions on storage theory.

<sup>13</sup>Deterministic components were cross-checked by visual inspection of the log-level series.

<sup>14</sup><https://cbspeeches.com>

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## Appendix A

This appendix documents structural breaks in the commodity price series underlying Figure 1 (Section 2.1). Monthly commodity prices (energy, agriculture, metals, and selected individual contracts) are from the World Bank Pink Sheet. All prices are deflated by the US CPI (FRED) and rebased to 1960 = 100. The sample spans 1960:01–2025:05 to match the figure. Additional monthly covariates used below are the effective federal funds rate (FFR, FRED), the real US dollar exchange rate (2020 = 100, FRED), and the NY Fed Global Supply Chain Pressure Index (GSCPI).

Annualized returns are 12-month log differences of the real price series,

$$\Delta_{12} \log X_t = \log X_t - \log X_{t-12}, \quad (1)$$

reported in percent as  $100 \times \Delta_{12} \log X_t$ . For each series that behaves as a unit-root process, we begin by estimating a single endogenous break in the deterministic component using the Perron (1989) procedure on log levels, allowing for a break in the intercept and/or the trend as appropriate. Table 4 reports the estimated break date and the selected deterministic specification.<sup>13</sup>

To allow for regime changes in the short-run mapping from macro covariates to commodity returns, we estimate multiple structural breaks in univariate regressions of the 12-month real return,  $C_t \equiv 100 \times \Delta_{12} \log X_t$  (Equation 1), on lagged policy and external variables. For each series, we consider three specifications:

$$\text{ModelA : } C_t = \alpha_1 + \alpha_2 \text{FFR}_{t-1} + \varepsilon_t, \quad (2)$$

$$\text{ModelB : } C_t = \alpha_1 + \alpha_2 \text{FFR}_{t-1} + \alpha_3 \text{RXR}_{t-1} + \varepsilon_t, \quad (3)$$

$$\text{ModelC : } C_t = \alpha_1 + \alpha_2 \text{FFR}_{t-1} + \alpha_3 \text{RXR}_{t-1} + \alpha_4 \text{GSCPI}_{t-1} + \varepsilon_t, \quad (4)$$

where FFR is the effective federal funds rate, RXR the real US dollar exchange rate (2020 = 100), and GSCPI the NY Fed Global Supply Chain Pressure Index. All covariates enter with one-month lags to limit simultaneity. Model A is the standard Frankel (1986)-type interest-rate link for storable commodities. Models B and C progressively condition on external drivers. Model B adds the real dollar,  $\text{RXR}_{t-1}$ , to capture the dollar-pricing/terms-of-trade channel, whereas Model C further includes  $\text{GSCPI}_{t-1}$  to control for time-varying supply bottlenecks. We report results using the nominal FFR and verify robustness with a proxy real rate  $\text{FFR} - \pi_{t-1}^{\text{PCE}}$  (break timing is qualitatively similar).

Break dates are selected using the Bai-Perron sequential sup  $F(L+1|L)$  procedure with a 10% trimming fraction, allowing up to three breaks, and a 5% significance level, unless noted (see Table 5). Series-specific exceptions are: soybeans (5% trim; 1% level) as well as wheat and copper (5% trim; 1% level; at most one break).

**TABLE 4** | Perron (1989) single-break tests on log levels.

| Series      | Break date | Deterministic component |
|-------------|------------|-------------------------|
| Crude oil   | 1972–12    | Constant                |
| Natural gas | 1973–12    | Constant                |
| Soybeans    | 2006–09    | Constant & Trend        |
| Corn        | 2006–06    | Constant & Trend        |
| Wheat       | 2005–07    | Constant & Trend        |
| Copper      | 1974–04    | Constant                |
| Gold        | 1980–01    | Constant & Trend        |

We also assess long-run comovement among the (log) real commodity price series using Johansen trace and maximum-eigenvalue tests. Across standard lag-length choices and deterministic specifications (constant in the VAR and in the cointegrating relations), the tests consistently indicate three cointegrating vectors ( $r=3$ ) over 1960:01–2025:05. This implies more than one common stochastic trend in the joint system, which complicates full-system modeling and motivates the univariate break analysis reported above.

## Appendix B

We use the CBSpeeches corpus compiled by Campiglio et al. (2025).<sup>14</sup> The dataset provides 35,487 full text speeches delivered between January 1986 and December 2023 by 131 monetary authorities worldwide. To compare communication across diverse monetary regimes and commodity exposures, a set of 14 central banks was selected, including the Bank of Canada, Bank of England, Bank of Japan, Bank of Mexico, Bank of Thailand, Board of Governors of the Federal Reserve, Central Bank of Chile, Central Bank of Kenya, Central Bank of the Philippines, European Central Bank, Monetary Authority of Singapore, Reserve Bank of Australia, Reserve Bank of India, and South African Reserve Bank.

After excluding all but these central banks from our sample, we retain a total of 13,643 speeches. Commodity references were detected through a single case-insensitive regular expression that concatenates the following tokens with the OR operator and is surrounded by word boundaries: commodity, commodities, raw material, oil, crude oil, natural gas, gasoline, energy, gold, silver, copper, iron ore, steel, metal, metals, wheat, corn, soybean, soybeans, rice, food price, and agriculture. The pattern, therefore, matches an occurrence only when the token appears as an independent lexical item (for instance, “oil” but not “spoiled”). The pattern flags either (i) the generic term commodity; (ii) specific energy, metal, or agricultural products; or (iii) the broader notions of energy and food prices.

Let  $m_i$  denote the number of matched tokens in speech  $i$ , and  $w_i$  its word count. The following statistics were computed:  $m_i$  (raw mentions per speech), an indicator  $1\{m_i > 0\}$ , and a normalized frequency  $10^3 m_i / w_i$  (mentions per 1 000 words, with zero if  $w_i = 0$ ). Aggregating by central bank yields (i) the total number of speeches, (ii) the share of speeches that contain at least one commodity term, (iii) the mean number of mentions per speech, and (iv) the mean number of mentions per 1 000 words.

Table 6 summarizes the findings for the period 1986–2023. Three broad patterns emerge. First, commodity exporters (for example, Canada, South Africa, and Chile) devote a much larger share of their public communication to commodity topics than do large, diversified advanced economies (such as the European Central Bank and the Bank of England) or service-oriented financial centers (like the Monetary Authority of Singapore). Second, the average intensity of discussion, measured either as raw hits or normalized by speech length, broadly mirrors the extensive margin: authorities that speak about commodities often also do so at greater length. Third, commodity importers with significant pass-through to inflation (Reserve Bank of India, Bank of Japan) exhibit high absolute mention counts, although the normalized intensity is more moderate, owing to the generally longer format of their speeches.

The three broad regularities discussed above are reinforced by additional evidence on (i) the overall length of speeches, (ii) the conditional distribution of commodity intensity, and (iii) the organizational level of the speaker. Table 7 pools the first two sets of statistics, while Table 8 reports the role composition of commodity remarks.

**Speech Length.** Median speeches by the Bank of England contain almost 4000 words, and the 90th-percentile speech exceeds 8100 words. The single longest intervention in the sample is a 65,479-word transcript by the Federal Reserve. By contrast, commodity exporters such as Canada and South Africa typically issue shorter addresses (medians below 2900 words). Length, therefore, attenuates the measured intensity for the Bank of England and, to a lesser extent, the Federal Reserve and the European Central Bank.

**Intensity Distribution.** Exporters again stand out: The median Bank of Canada speech already contains 1.22 commodity mentions per 1000

**TABLE 5** | Break dates by specification (Bai–Perron multiple breaks).

| Series      | Model A                   | Model B                   | Model C                   |
|-------------|---------------------------|---------------------------|---------------------------|
| Crude oil   | 1973–10, 1980–03          | 1975–01, 1999–04          | None                      |
| Natural gas | None                      | None                      | 2002–04, 2021–01          |
| Soybeans    | None                      | 1974–01, 2007–06, 2015–07 | 2004–09, 2006–11, 2015–02 |
| Corn        | 2006–09, 2018–04          | 1974–05, 2005–10, 2013–07 | 2002–05, 2005–07, 2013–07 |
| Wheat       | 2007–09, 2017–07          | 1974–05, 2006–02, 2017–05 | 2005–08                   |
| Copper      | 1967–06, 1974–06          | 1970–06                   | 2000–11, 2007–05, 2020–02 |
| Gold        | 1971–01, 1980–12, 2012–05 | 1974–05, 1980–12, 2012–07 | 2012–05                   |

Note. Dates are YYYY–MM. “None” indicates no statistically significant break at the stated trimming and significance levels used for each model.

**TABLE 6** | Commodity references in speeches, 1986–2023.

| Central bank                              | Share of speeches<br>with commodity<br>reference (%) | Mean hits per<br>speech | Mean hits per<br>1 000 words | Notable observations                                                                                                  |
|-------------------------------------------|------------------------------------------------------|-------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Bank of Canada                            | 74.8                                                 | 5.58                    | 2.44                         | Large exporter of energy, metals, and agricultural goods; terms of trade and inflation pressures feature prominently. |
| South African Reserve Bank                | 73.1                                                 | 5.44                    | 1.83                         | Gold, platinum-group metals, and coal prices are frequently linked to growth, the external balance, and the rand.     |
| Bank of Japan                             | 72.6                                                 | 5.94                    | 1.48                         | Energy and metal import costs are treated as key drivers of inflation expectations.                                   |
| Central Bank of Chile                     | 69.3                                                 | 6.62                    | 1.73                         | Copper prices are central to macroeconomic dynamics and receive extensive attention.                                  |
| Reserve Bank of India                     | 61.1                                                 | 5.56                    | 1.37                         | Imported crude and edible oils dominate discussions of headline inflation.                                            |
| Bank of England                           | 54.5                                                 | 4.02                    | 0.90                         | Commodity talk rises in episodes of energy-price shocks, but is generally concise.                                    |
| Reserve Bank of Australia                 | 54.4                                                 | 2.91                    | 0.89                         | Despite Australia’s export profile, shorter bulletin-style pieces reduce the intensity measure.                       |
| Bank of Mexico                            | 51.1                                                 | 2.81                    | 1.38                         | Oil-price developments are linked to both inflation and fiscal revenues.                                              |
| Bank of Thailand                          | 51.0                                                 | 2.40                    | 0.95                         | Agricultural and energy items underpin headline inflation commentary.                                                 |
| Board of Governors of the Federal Reserve | 45.8                                                 | 3.27                    | 1.03                         | References arise mainly in inflation and financial-stability contexts.                                                |
| European Central Bank                     | 42.8                                                 | 2.06                    | 0.64                         | Discussion concentrates on specific energy shocks, reflecting a diversified economic structure.                       |
| Monetary Authority of Singapore           | 37.0                                                 | 1.64                    | 0.83                         | Mentions are episodic and relate to imported inflation pressures.                                                     |
| Central Bank of the Philippines           | 41.5                                                 | 1.20                    | 0.79                         | Commodity discourse centres on food and energy shocks.                                                                |
| Central Bank of Kenya                     | 27.6                                                 | 1.05                    | 1.14                         | The bank refers to external energy and food shocks less frequently but at a slightly higher density when it does.     |

**TABLE 7** | Speech length and conditional intensity, 1986–2023.

| Central bank                    | Speech length (words) |           |         | Mentions per 1000 words |          |          |
|---------------------------------|-----------------------|-----------|---------|-------------------------|----------|----------|
|                                 | Median                | 90th pct. | Maximum | $P_{50}$                | $P_{75}$ | $P_{90}$ |
| Bank of Canada                  | 2627                  | 3978      | 11 888  | 1.22                    | 3.02     | 6.11     |
| Bank of England                 | 3957                  | 8151      | 26 285  | 0.17                    | 0.75     | 2.28     |
| Bank of Japan                   | 4244                  | 6778      | 14 166  | 0.79                    | 1.88     | 3.85     |
| Bank of Mexico                  | 1888                  | 3109      | 7860    | 0.28                    | 1.55     | 3.38     |
| Bank of Thailand                | 2036                  | 3566      | 6 584   | 0.24                    | 1.04     | 2.54     |
| Board of Governors of the Fed   | 3051                  | 5030      | 65 479  | 0.00                    | 0.72     | 2.24     |
| Central Bank of Chile           | 2699                  | 4309      | 9308    | 0.98                    | 2.52     | 5.03     |
| Central Bank of Kenya           | 1533                  | 2739      | 4694    | 0.00                    | 1.14     | 2.96     |
| Central Bank of the Philippines | 1328                  | 2329      | 4694    | 0.00                    | 0.97     | 2.22     |
| European Central Bank           | 2770                  | 5208      | 19 639  | 0.00                    | 0.60     | 1.76     |
| Monetary Authority of Singapore | 1865                  | 3323      | 6532    | 0.00                    | 0.63     | 2.09     |
| Reserve Bank of Australia       | 3269                  | 4625      | 10 036  | 0.26                    | 0.95     | 2.44     |
| Reserve Bank of India           | 3519                  | 6397      | 17 953  | 0.36                    | 1.24     | 3.21     |
| South African Reserve Bank      | 2826                  | 4058      | 7910    | 0.98                    | 2.45     | 4.61     |

**TABLE 8** | Share of commodity mentions by speaker role.

| Central bank                    | Governor | Deputy governor | Board or committee member | Senior management |
|---------------------------------|----------|-----------------|---------------------------|-------------------|
| Bank of Canada                  | 50.0     | 16.0            | 33.0                      | 1.0               |
| Bank of England                 | 10.2     | 23.1            | 54.0                      | 12.7              |
| Bank of Japan                   | 33.2     | 21.3            | 45.3                      | 0.1               |
| Bank of Mexico                  | 37.1     | 62.9            | 0.0                       | 0.0               |
| Bank of Thailand                | 90.5     | 9.0             | 0.0                       | 0.0               |
| Board of Governors of the Fed   | 43.0     | 13.3            | 43.6                      | 0.1               |
| Central Bank of Chile           | 45.9     | 30.5            | 23.6                      | 0.0               |
| Central Bank of Kenya           | 66.0     | 30.5            | 0.0                       | 3.5               |
| Central Bank of the Philippines | 99.2     | 0.8             | 0.0                       | 0.0               |
| European Central Bank           | 30.5     | 12.5            | 57.0                      | 0.0               |
| Monetary Authority of Singapore | 5.2      | 14.6            | 9.2                       | 71.0              |
| Reserve Bank of Australia       | 45.9     | 26.5            | 0.0                       | 28.0              |
| Reserve Bank of India           | 33.3     | 54.7            | 0.0                       | 12.0              |
| South African Reserve Bank      | 66.0     | 30.5            | 0.0                       | 3.5               |

words, and the top decile reaches 6.11. The South African Reserve Bank exhibits a similar but slightly lighter upper tail. In diversified or importer economies (Federal Reserve, European Central Bank, Bank of England), the distribution is bimodal: half of the speeches remain silent, whereas the upper quartile and above become dense only during episodes of commodity-price turbulence.

**Speaker Hierarchy.** Commodity communication is frequently handled at the top of the institutional hierarchy when it is macro-critical. Governors account for one-half of mentions at the Bank of Canada and two-thirds at the South African Reserve Bank. Where responsibility is more diffuse, either due to committee governance (e.g., the Bank of England, European Central Bank) or portfolio assignments (e.g., the Reserve Bank of India), deputy governors or ordinary board members

tend to dominate. The Monetary Authority of Singapore and the Reserve Bank of Australia exemplify a third model, in which senior managerial staff contribute a substantial share of statements, reflecting an analytical, data-driven approach.

**Market-Driven Relationship?** Speech metrics were aggregated to monthly frequency and merged with World Bank price returns for seven benchmark commodities. For each central bank  $c$ , we compute the vector of Pearson correlations

$$\rho_{c,k} = \text{corr}(d_{c,t}, r_{k,t}), \quad k \in \{\text{oil, gas, soybeans, corn, wheat, copper, and gold}\},$$

where  $r_{k,t}$  is the month-ahead percentage change in commodity  $k$ . The salient findings are:

**TABLE 9** | Central bank speeches that discuss commodities.

| Speaker            | Central Bank                              | Date       | # KW | Title                                                                                           |
|--------------------|-------------------------------------------|------------|------|-------------------------------------------------------------------------------------------------|
| Fabio Panetta      | European Central Bank                     | 2022-11-16 | 180  | Greener and Cheaper: Could the Transition Away from Fossil Fuels Generate a Divine Coincidence? |
| Andrew Bailey      | Bank of England                           | 2023-11-20 | 144  | A measure of wheat for a penny– food price inflation in historical perspective                  |
| Timothy Lane       | Bank of Canada                            | 2015-01-13 | 119  | Drilling down– understanding oil prices and their economic impact                               |
| Swati Dhingra      | Bank of England                           | 2023-03-08 | 117  | A cost-of-living crisis: Inflation during an unprecedented terms-of-trade shock                 |
| Silvana Tenreyro   | Bank of England                           | 2022-11-11 | 88   | The path to 2%                                                                                  |
| Jon Cunliffe       | Bank of England                           | 2022-04-04 | 88   | Monetary and financial stability and the invasion of Ukraine                                    |
| Isabel Schnabel    | European Central Bank                     | 2020-02-27 | 88   | How long is the medium term? Monetary policy in a low-inflation environment                     |
| Daniel K. Tarullo  | Board of Governors of the Federal Reserve | 2014-11-21 | 84   | Physical commodities                                                                            |
| Harun R. Khan      | Reserve Bank of India                     | 2015-02-23 | 78   | Emerging trends in the global financial landscape– India's challenges and opportunities         |
| Isabel Schnabel    | European Central Bank                     | 2022-01-08 | 77   | Looking through higher energy prices? Monetary policy and the green transition                  |
| Haruhiko Kuroda    | Bank of Japan                             | 2022-04-22 | 76   | Japan's Inflation Dynamics and the Role of Monetary Policy                                      |
| Rodrigo Vergara    | Central Bank of Chile                     | 2014-10-23 | 73   | Commodity Prices and Macroeconomic Policy                                                       |
| Lynn Patterson     | Bank of Canada                            | 2016-03-30 | 68   | Adjusting to the fall in commodity prices– one step at a time                                   |
| Lawrence Schembri  | Bank of Canada                            | 2016-11-08 | 66   | Wood, wheat, wheels and the web – historical pivots and future prospects for Canadian exports   |
| Michelle W. Bowman | Board of Governors of the Federal Reserve | 2019-03-28 | 66   | Agriculture and community banking                                                               |
| Mahesh Kumar Jain  | Reserve Bank of India                     | 2019-11-13 | 66   | Rural and agriculture finance– critical input to achieve inclusive and sustainable development  |
| Jonathan Haskel    | Bank of England                           | 2023-11-28 | 63   | UK inflation since the pandemic: How did we get here and where are we going?                    |
| Jonathan Haskel    | Bank of England                           | 2022-11-11 | 63   | Recent UK monetary policy in a changing economy                                                 |
| Philip R. Lane     | European Central Bank                     | 2023-03-06 | 61   | Underlying inflation                                                                            |
| Masazumi Wakatabe  | Bank of Japan                             | 2022-06-01 | 58   | Japan's economy and monetary policy                                                             |
| Isabel Schnabel    | European Central Bank                     | 2022-03-17 | 58   | A new age of energy inflation: climateflation, fossilflation and greenflation                   |
| Ravi Menon         | Monetary Authority of Singapore           | 2022-08-17 | 57   | What does it take to get to net zero                                                            |

(Continues)



TABLE 9 | (Continued)

| Speaker          | Central Bank               | Date       | # KW | Title                                                          |
|------------------|----------------------------|------------|------|----------------------------------------------------------------|
| Francois Groepe  | South African Reserve Bank | 2015-01-29 | 55   | Recent developments in the African region                      |
| Gertjan Vlieghe  | Bank of England            | 2017-09-15 | 55   | Real interest rates and risk                                   |
| Stephen S. Poloz | Bank of Canada             | 2015-09-21 | 55   | Riding the commodity cycle– resources and the Canadian economy |

Note. “# KW” = keyword count.

- **Coefficients cluster tightly around zero.** Across 98 central-bank-commodity pairs, 90% of correlations lie within the interval  $[-0.10, 0.10]$ . Monthly variation in commodity communication is therefore only weakly (and noisily) aligned with contemporaneous price moves.
- **Signals with structural intuition are present but small.** The RBI shows the largest positive coefficients vis-à-vis natural gas and copper returns ( $\rho \approx 0.15$ ), consistent with its import dependence; the BoC records the strongest positive link with gold ( $\rho = 0.12$ ), reflecting Canada’s sizeable bullion sector. Yet the magnitudes remain economically negligible.
- **Negative coefficients dominate some exporters.** Energy and metal exporters such as the BoC and SARB post weakly negative correlations (e.g., BoC vs. natural gas,  $-0.09$ ), a result that plausibly reflects base effects or lags in the speech-production process rather than counter-cyclical messaging.
- **The ECB remains flat.** All ECB coefficients are  $|\rho| \leq 0.06$ , reinforcing the view that contemporaneous commodity fluctuations rarely permeate euro-area discourse.

This evidence suggests that central banks acknowledge commodity shocks, albeit in an episodic and nonlinear manner, rather than through a mechanical month-by-month response. Structural exporters and importers show faint but directionally coherent sensitivities, suggesting that real-sector exposure informs the tone of communication. Nonetheless, the contemporaneous market signal is dwarfed by the influence of the institutional communication calendar and the inherent lags in speech preparation.

**Lagged Correlations.** The weak contemporaneous links documented could in principle mask (i) a reaction of central-bank language to recent commodity shocks or (ii) a forward-looking content that anticipates subsequent market moves. To address these possibilities, the return vectors were shifted by  $L \in \{-3, -2, -1, 0, 1, 2, 3\}$  months, and the correlation matrix

$$\rho_{c,k}(L) = \text{corr}(d_{c,t}, r_{k,t-L})$$

was recomputed for every bank  $c$  and commodity  $k$ .

Across 686 coefficient estimates, 92% fall within the interval  $[-0.20, 0.20]$  and 70% within  $[-0.10, 0.10]$ . Lead-lag transformations, therefore, do not alter the headline result that linear dependence between monthly speech density and individual commodity returns is economically trivial, coefficients at negative lags, where speeches would Granger-cause subsequent returns, remain essentially random. In sum, our analysis indicates that the observed language is neither distinctly reactive nor predictive in a linear sense at this frequency. Consequently, the earlier conclusion that the communication calendar, editorial lead times, and threshold behavior dominate contemporaneous speech dynamics is robust to alternative timing assumptions.

Table 9 lists the 20 highest-commodity-content speeches (measured by keyword count), along with the speaker, central bank, date, and title.